

On the Prediction Error and Excess Risk Bounds

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Hamburg, 22.07.2024

A handwritten signature in black ink, appearing to read 'A. Ludwig'.

Abstract

Regularized least-squares estimation is a powerful tool to leverage sparsity in high-dimensional models and to prevent overfitting. We will present an oracle bound on the prediction error under the assumptions that the underlying function class from which the estimation is performed is convex, and that the regularizing term is a squared semi-norm on this function class. The result is versatile in that it yields optimal rates for parametric and non-parametric models. We then switch the setting from fixed to random covariates and illustrate how uniform laws can be used to yield excess risk bounds from prediction error bounds. The uniform law is of interest by itself, since it shows how theory from concentration inequalities of empirical processes can be utilized in the context of bounding risk. We then look at refinements of uniform laws and aim to obtain sharper uniform laws by further constraining the underlying function class.

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Chapter 1

Introduction

Regression analysis provides a set of tools and procedures to estimate the relationship between a *response variable* and one or more *covariates* by modeling the response variable as a function of the covariates. In the true underlying model, the response variable is sampled as a function of the covariates and an additional zero-mean *noise* variable. This function, known as the *regression function* minimizes the *mean-squared error (MSE)*. Calculating this function is not possible without knowing the joint distribution of covariates and response variable, so a procedure for estimating this function is necessary. Given observed data points in the form of response variable and covariate pairs, the process of modeling their relationship involves two key components: 1. an appropriate function class to estimate from, and 2. the procedure used to determine the estimate within that class. In the non-parametric setting little assumptions are made about the statistical model and usually, it is infinite-dimensional [21].

A common way to measure the accuracy of the estimated function is assessing the *excess risk*, meaning the difference between the MSE associated with the estimate at hand and the optimal MSE associated with the regression function. Assuming fixed covariates and moving to an empirical analog of the MSE, namely the *empirical mean-squared error (empirical MSE)* one could choose the function in the underlying function class to minimize the empirical MSE. This procedure is known as *least-squares estimation (LSE)*, which has established itself as a powerful tool in applications. Under mild assumptions on the distribution of the noise, this estimate also presents attractive theoretical properties. For the method of LSE bounds on the *prediction error*, the empirical analog of excess risk, can be derived by reducing the problem to finding tail-bounds on the supremum of a stochastic process indexed by the underlying function class [17, 20]. Tools from the theory of concentration inequalities [4, 19] yield said bounds, This approach yields optimal rates for certain choices of function classes in both the parametric and non-parametric setting [20, ch. 13 & 15].

In the parametric case of linear regression, the prediction error takes the form $\mathcal{O}(d/n)$, where n denotes the sample size, and d the dimension of the covariates [11, p. 9]. In classical statistics typically the model is assessed under the

assumption that n diverges, while d remains fixed, which would provide valuable insight on the performance of the LSE here. In high-dimensional models (d and n large and $d \approx n$ or even $d \gg n$), the bound on the prediction error does not indicate satisfactory performance of the LSE and rather illustrates difficulties that arise in high-dimensional models. However, in many practical applications, the true underlying model is sparse, meaning that only a relatively small number s of the covariates are relevant for predicting the response variable. Using the LSE in such a setting can result in overfitted estimate, meaning an estimate that overinterprets the underlying noise and misses the essential structure of the model. Sparsity allows for improved bounds on the prediction error because it reduces the effective dimensionality of the problem [1, 5, 10]. Techniques that induce sparsity in the estimate like the lasso estimator leverage sparsity by adding an ℓ_1 -regularizing term to the least-squares loss function, thereby improving the accuracy of the model [7, 15, 16]. Under suitable conditions, combining power two bounds [11, ch. 6] and ideas from Bellec and Tsybakov [2], the lasso yields a prediction error bound $\mathcal{O}(s \log(d)/n)$, which depends linearly on the number of true explanatory covariates and only grows logarithmically on the total dimension of the covariates. In the nonparametric setting usually instead of sparsity, smoothness assumptions are made.

Changing the setting from fixed covariates to covariates being sampled from an underlying distribution, finding bounds on the excess risk as opposed to the prediction error becomes of interest. Further tools from the theory of concentration inequalities [4] allow a connection between bounds on the prediction error and bounds on the excess risk [20].

In this thesis, we first establish a bound on the prediction error for a regularized least-squares estimator, incorporating a squared semi-norm on the function class [20, Theorem 13.17]. Following this, we prove a one-sided uniform law [20, Theorem 14.12], which enables us to deduce a bound on the excess risk. Finally, we delve into refinements of the one-sided uniform law.

Chapter 2 introduces the problem set-up, various measures of estimator accuracy, and different estimation procedures. In Chapter 3, we provide an overview of basic concentration inequalities, specifically tail-bounds for sub-Gaussian random variables. Chapter 4 introduces local Gaussian complexity and explains how critical inequalities can link the concentration inequalities discussed in Chapter 3 to prediction error bounds. In Chapter 5, we present a proof of an oracle bound on the prediction error for the regularized least-squares estimate and demonstrate its application. Chapter 6 transitions to the setting of variable covariates and presents a one-sided uniform law that connects prediction error bounds to excess risk bounds. We then explore further refinements of the one-sided uniform law.

Chapter 2

Problem Set-up

In this chapter, we present the general problem set-up for regression. We discuss how the quality of an estimate can be assessed, which procedures can be used for estimation, and a short preview of the approach to bound the prediction error.

2.1 Different measures of quality

Given a real-valued random vector X that takes value in $\mathcal{X} \subset \mathbb{R}^d$ and a real-valued random variable Y that takes values in $\mathcal{Y}$, in the *random design* version of regression the aim is to find a function $f : \mathcal{X} \rightarrow \mathcal{Y}$ to minimize the so-called *mean-squared error* (MSE)

$$\mathcal{L}_f := E_{X,Y}[(Y - f(X))^2].$$

We call X *covariates* and Y the *response variable*. If relevant expectations exist the minimizing function is given by the *Bayes' least squares estimate* or *regression function* $f^*(\mathbf{x}) = E[Y|X = \mathbf{x}]$, see [8, p. 213] or [3, ch. 34]. However, the joint distribution of (X, Y) is not known a-priori, so it is not viable to explicitly calculate f^* . Instead we turn to an empirical analog: Given pairs $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ drawn from (X, Y) one can compute the empirical mean-squared error

$$\hat{\mathcal{L}}_f := \frac{1}{n} \sum_{i=1}^n (y_i - f(\mathbf{x}_i))^2.$$

The method of *nonparametric least squares* is based on minimizing the empirical mean-squared error over a suitably chosen function class.

Given any estimate f a natural way to measure the quality of the estimate is to calculate the difference between the MSE $\mathcal{L}_f$ and the optimal MSE $\mathcal{L}_{f^*}$. The so-called *excess risk*, takes the form

$$\mathcal{L}_f - \mathcal{L}_{f^*} = E_X[(f(X) - f^*(X))^2],$$

since

$$\begin{aligned}
\mathcal{L}_f - \mathcal{L}_{f^*} &= E_X[(Y - f(X))^2] - E_X[(Y - f^*(X))^2] \\
&= -2E_X[Y(f(X) - f^*(X))] + E_X[f^2(X)] - E_X[(f^*)^2(X)] \\
&\stackrel{i)}{=} -2E_X[E_X[Y(f(X) - f^*(X))|X]] + E_X[f^2(X)] - E_X[(f^*)^2(X)] \\
&\stackrel{ii)}{=} -2E_X[E_X[Y|X](f(X) - f^*(X))] + E_X[f^2(X)] - E_X[(f^*)^2(X)] \\
&\stackrel{iii)}{=} E_X[(f(X) - f^*(X))^2]
\end{aligned}$$

where we use the tower property of the conditional expectation in *i*), the fact that $E[Yg(X)|X] = g(X)E[Y|X]$ for a (measurable) function g , in *ii*) and the fact that $E[Y|X] = f^*(X)$ in *iii*). For a complete revision of these properties see [3, ch. 34]. Let P denote the distribution of X , then defining the $L^2(P)$ -semi-norm as

$$\|g\|_{L^2(P)} := \left[\int g^2(\mathbf{x}) P(d\mathbf{x}) \right]^{1/2}$$

for any function g such that the right-hand integral is finite¹, we then denote $g \in L^2(P)$ and also introduce the shorthand notation $\|g\|_2 := \|g\|_{L^2(P)}$. Then by a transformation of variables we obtain

$$E_X[(f(X) - f^*(X))^2] = \|f - f^*\|_2^2.$$

To calculate this quantity f^* needs to be known and hence we turn to an empirical analog again:

Let $\{\mathbf{x}_i\}_{i=1}^n$ be n samples from the covariates, so $\mathbf{x}_i \in \mathcal{X}$ for $i = 1, \dots, n$. We define the empirical distribution $P_n := \frac{1}{n} \sum_{i=1}^n \delta_{\mathbf{x}_i}$ where δ_x is the indicator function of $\{x\}$, then analogously we define the $L^2(P_n)$ -semi-norm to be

$$\|g\|_{L^2(P_n)} := \left[\frac{1}{n} \int g^2(\mathbf{x}) P_n(d\mathbf{x}) \right]^{1/2} = \left[\frac{1}{n} \sum_{i=1}^n g(\mathbf{x}_i)^2 \right]^{1/2}$$

for any function $g : \mathcal{X} \rightarrow \mathcal{Y}$ where the second inequality follows by definition of the integral and P_n . We introduce the shorthand notation $\|\cdot\|_n := \|\cdot\|_{L^2(P_n)}$. The associated *prediction error* is given by

$$\|f - f^*\|_n = \left[\frac{1}{n} \sum_{i=1}^n (f(\mathbf{x}_i) - f^*(\mathbf{x}_i))^2 \right]^{1/2}.$$

Viewing the samples $\{\mathbf{x}_i\}_{i=1}^n$ as fixed is a set-up known as regression with *fixed design*. It remains to discuss how one chooses this estimate f .

¹We omit a rigorous measure theoretic definition to avoid unnecessary digressions, see [3, p.241-243] for details

2.2 Choosing the estimate

For fixed design points $\{\mathbf{x}_i\}_{i=1}^n$ in $\mathcal{X}$ by properties of the conditional expectation we can write

$$y_i = f^*(\mathbf{x}_i) + v_i, \quad \text{for } i = 1, \dots, n, \quad (2.2.1)$$

where v_i is a zero-mean random variable. In the *standard nonparametric regression model* we assume that $v_i = \sigma w_i$ where w_i are drawn i.i.d. from a $\mathcal{N}(0, 1)$ distribution, so the v_i are drawn i.i.d. from a $\mathcal{N}(0, \sigma^2)$ distribution. One way in which to estimate the regression function is to solve the problem

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \left\{ \frac{1}{n} \sum_{i=1}^n (y_i - f(\mathbf{x}_i))^2 \right\} \quad (2.2.2)$$

over some chosen function class $\mathcal{F}$ where all members of this function class have the common domain $\mathcal{X}$. It can also be convenient to introduce a regularizing term with respect to a norm $\|\cdot\|_{\mathcal{F}}$ on the given function class $\mathcal{F}$, see Chapter 1, and solve the problem:

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \left\{ \frac{1}{2n} \sum_{i=1}^n (y_i - f(\mathbf{x}_i))^2 + \lambda_n \|f\|_{\mathcal{F}}^2 \right\}, \quad (2.2.3)$$

where $\lambda_n \in (0, \infty)$ is the regularization weight. This will be the estimate to be analyzed in this text, although the techniques to be discussed can be applied to a variety of estimates, see [20, ch. 13 & 14] and [17, ch. 12].

2.3 Bounding prediction error

We aim to bound the prediction error $\|\hat{f} - f^*\|_n$, where $\hat{f}$ is a solution to problem (2.2.3). The starting point of our analysis is the following equation, which holds by definition on $\hat{f}$ and for all $\tilde{f} \in \mathcal{F}$:

$$\frac{1}{2n} \sum_{i=1}^n (y_i - \hat{f}(\mathbf{x}_i))^2 + \lambda_n \|\hat{f}\|_{\mathcal{F}}^2 \leq \frac{1}{2n} \sum_{i=1}^n (y_i - \tilde{f}(\mathbf{x}_i))^2 + \lambda_n \|\tilde{f}\|_{\mathcal{F}}^2. \quad (2.3.1)$$

We now want to introduce terms and rearrange the inequality to bound the prediction error: For ease of notation we introduce the terms $\hat{\Delta} := \hat{f} - f^*$ and $\tilde{\Delta} := \tilde{f} - f^*$. Replacing y_i by $f^*(\mathbf{x}_i) + \sigma w_i$ and using the newly introduced notation yields

$$\frac{1}{2n} \sum_{i=1}^n (\sigma w_i - \hat{\Delta}(\mathbf{x}_i))^2 + \lambda_n \|\hat{f}\|_{\mathcal{F}}^2 \leq \frac{1}{2n} \sum_{i=1}^n ((f^* - \tilde{f})(\mathbf{x}_i) + \sigma w_i)^2 + \lambda_n \|\tilde{f}\|_{\mathcal{F}}^2$$

and using the binomial formula, cancelling the term $\sum_{i=1}^n (\sigma w_i)^2 / n$ yields

$$\frac{1}{2} \|\hat{\Delta}\|_n^2 - \frac{\sigma}{n} \sum_{i=1}^n \hat{\Delta}(\mathbf{x}_i) w_i \leq \frac{1}{2} \|f^* - \tilde{f}\|_n^2 + \frac{\sigma}{n} \sum_{i=1}^n (f^* - \tilde{f})(\mathbf{x}_i) w_i + \lambda_n (\|\tilde{f}\|_{\mathcal{F}}^2 - \|\hat{f}\|_{\mathcal{F}}^2)$$

now inserting $f^* - \tilde{f} = \tilde{\Delta} - \hat{\Delta}$ and cancelling the term $\sum_{i=1}^n \hat{\Delta}(\mathbf{x}_i) \sigma w_i / n$ yields

$$\frac{1}{2} \|\hat{\Delta}\|_n^2 \leq \frac{1}{2} \|\tilde{f} - f^*\|_n^2 + \frac{\sigma}{n} \sum_{i=1}^n w_i \tilde{\Delta}(\mathbf{x}_i) + \lambda_n (\|\tilde{f}\|_{\mathcal{F}}^2 - \|\hat{f}\|_{\mathcal{F}}^2). \quad (2.3.2)$$

For our purposes, we will deal with the possibly weaker bound²

$$\frac{1}{2} \|\hat{\Delta}\|_n^2 \leq \frac{1}{2} \|\tilde{f} - f^*\|_n^2 + \frac{\sigma}{n} \left| \sum_{i=1}^n w_i \tilde{\Delta}(\mathbf{x}_i) \right| + \lambda_n (\|\tilde{f}\|_{\mathcal{F}}^2 - \|\hat{f}\|_{\mathcal{F}}^2). \quad (2.3.3)$$

If it is possible to further upper bound the right-hand side of the equation by a term of the form $a \|\hat{\Delta}\|_n + b$, where a, b are constants that do not depend on $\hat{f}$ but only on f^* and properties of the function class $\mathcal{F}$, then it is straightforward to find a bound on the prediction error $\|\hat{\Delta}\|_n$ by the quadratic formula³. To do so it will be essential to control the random variable $|\sum_{i=1}^n w_i \tilde{\Delta}(\mathbf{x}_i)|$ in a uniform manner.

2.4 A short note on measurability

In the following sections we will often deal with variables whose measurability is dependent on some underlying function class $\mathcal{F}$ (not necessarily the same function class as in the previous discussion). More specifically we will deal with variables Z indexed by functions of the function class $\mathcal{F}$ (in some cases this can be a subset of $\mathbb{R}^n$), we denote this by Z_f . In all the following cases Z_f is measurable if f is measurable and we do assume that all members $f \in \mathcal{F}$ are measurable. However, the variable

$$\sup_{f \in \mathcal{F}} Z_f$$

is not necessarily measurable. To still make statements about the quantities resembling distributions and expected values we adopt the following convention [18]: For any measurable Φ

$$E \left[\Phi \left(\sup_{f \in \mathcal{F}} Z_f \right) \right] := \sup \left\{ E \left[\Phi \left(\sup_{f \in \mathcal{H}} Z_f \right) \right] \mid \mathcal{H} \subset \mathcal{F} \text{ countable} \right\}. \quad (2.4.1)$$

Note that the expectations on the right-hand side are well defined since a countable supremum of measurable functions is measurable. We will not further comment on issues of measurability unless necessary and refer the reader to [18] for more details.

²Note that if we uniformly bound the right side for all possible values $\tilde{\Delta}$ can take, then this bound is not weaker. $\tilde{\Delta}$ can take the form of any function in $\partial\mathcal{F} := \{f_1 - f_2 \mid f_1, f_2 \in \mathcal{F}\}$, which is symmetric, meaning $g \in \partial\mathcal{F}$ implies $-g \in \partial\mathcal{F}$.

³Assuming the constants a, b permit zeros of the polynomial

Chapter 3

Basic Concentration Inequalities

In this chapter, we introduce the basic concentration inequalities for sub-Gaussian random variables which are utilized as mathematical tools to bound the prediction error. We also introduce the concept of Gaussian complexity, a quantity that characterizes the size of a subset of $\mathbb{R}^n$.

3.1 Sub-Gaussian property

Essentially we want to define classes of real-valued random variables such that we can bound the quantity $P(X - E[X] \geq t)$ as a function of $t \geq 0$. The moment-generating function $E[e^{\lambda X}]$ for λ ranging in some connected subset of $\mathbb{R}$ in which it exists is a valuable tool since we can use Markov's inequality:

$$P(X - \mu \geq t) = P(e^{\lambda(X-\mu)} \geq e^{\lambda t}) \leq \frac{E[e^{\lambda(X-\mu)}]}{e^{\lambda t}}.$$

We can then optimize for λ to find a sharp bound.

The following example is central:

Example 3.1.1. Let $X \sim \mathcal{N}(\mu, \sigma^2)$, then one has $E[e^{\lambda(X-\mu)}] = e^{\frac{\lambda^2 \sigma^2}{2}}$ for all $\lambda \in \mathbb{R}$ [3, p. 278-280]. Using the discussion above we obtain

$$P(X - \mu \geq t) \leq e^{\frac{\lambda^2 \sigma^2}{2} - \lambda t}.$$

Optimizing for λ using elementary calculus yields

$$\min_{\lambda \in \mathbb{R}} \left\{ \frac{\lambda^2 \sigma^2}{2} - \lambda t \right\} = -\frac{t^2}{2\sigma^2},$$

where the argument $\lambda^* = t/\sigma^2$ maximizes the term. By the prior discussion this yields $P(X - \mu \geq t) \leq e^{-\frac{t^2}{2\sigma^2}}$.

The prior example motivates the following definition

Definition 3.1.1 (Sub-Gaussian random variables). A random variable X with mean μ is called sub-Gaussian if a parameter σ exists such that

$$E[e^{\lambda(X-\mu)}] \leq e^{\frac{\lambda^2 \sigma^2}{2}}.$$

For sub-Gaussian random variables the moment generating function exists for all $\lambda \in \mathbb{R}$ by definition, so we can also apply the same bounding technique to $-X$ to find bounds on $P(\mu - X \geq t)$ and thereby also bounds on $P(|X - \mu| \geq t)$. In example 3.1.1 we only used bounds on the moment generating function, so the following theorem immediately follows, see [20, ch. 2]:

Theorem 3.1.1 (Sub-Gaussian tail bounds). For a sub-Gaussian random variable X with parameter σ the following tail-bound holds

$$P(X - \mu \geq t) \leq e^{-\frac{t^2}{2\sigma^2}}$$

and

$$P(|X - \mu| \geq t) \leq 2e^{-\frac{t^2}{2\sigma^2}}.$$

Remark: These tail-bounds provide non-asymptotic control over deviations from the mean. Although asymptotic results are contained such as the weak law of large numbers, for $t \rightarrow \infty$, the rate contained in the parameter σ provides non-asymptotic insight on the concentration of X around its mean.

3.2 Lipschitz-functions of Gaussian variables

Remember the motivation to bound $|\sum_{i=1}^n w_i \tilde{\Delta}(\mathbf{x}_i)|$, this can be interpreted as a function of the vector $(w_1, \dots, w_n)$ and as it turns out this interpretation yields helpful bounds:

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is called *L-Lipschitz* with respect to euclidean norm $\|\cdot\|$ if

$$|f(\mathbf{x}) - f(\mathbf{y})| \leq L\|\mathbf{x} - \mathbf{y}\| \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

The following theorem allows us to control tail bounds of *L-Lipschitz* functions of Gaussian variables

Theorem 3.2.1. Let $\mathbf{w} = (w_1, \dots, w_n)$ be a vector of i.i.d. standard Gaussian variables, and let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be *L-Lipschitz* with respect to euclidean norm $\|\cdot\|_2$, then the variable $f(\mathbf{w}) - E[f(\mathbf{w})]$ is sub-Gaussian with parameter at least L , and hence

$$P(|f(\mathbf{w}) - E[f(\mathbf{w})]| \geq t) \leq 2e^{-\frac{t^2}{2L^2}} \quad \forall t \geq 0.$$

Proof. See [20, Th. 2.26] for a simple proof with weaker constants and [4, Th. 5.5] for a full proof. $\square$

An immediate and important application:

Example 3.2.1. Let $\mathcal{A} \subset \mathbb{R}^n$ be any bounded subset and $w = (w_1, \dots, w_n)$ be a vector of i.i.d. standard Gaussian variables, define the random variable

$$Z := \sup_{\mathbf{a} \in \mathcal{A}} \sum_{k=1}^n a_k w_k = \sup_{\mathbf{a} \in \mathcal{A}} \langle \mathbf{a}, w \rangle.$$

Now define $D(\mathcal{A}) := \sup_{\mathbf{a} \in \mathcal{A}} \|\mathbf{a}\|_2$. If we now treat Z as a function $(w_1, \dots, w_n) \mapsto f(w_1, \dots, w_n)$, we can show that f is L -Lipschitz with Lipschitz constant at most $D(\mathcal{A})$. Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\mathbf{a}^* \in \mathcal{A}$ chosen¹ such that $f(\mathbf{x}) = \langle \mathbf{a}^*, \mathbf{x} \rangle$, then

$$f(\mathbf{x}) - f(\mathbf{y}) \leq \langle \mathbf{a}^*, \mathbf{x} - \mathbf{y} \rangle \leq D(\mathcal{A}) \|\mathbf{x} - \mathbf{y}\|_2,$$

where the first inequality follows from $\langle \mathbf{a}^*, \mathbf{y} \rangle \leq f(\mathbf{y})$ by definition of f and second inequality follows by Cauchy Schwarz inequality. The same argumentation yields

$$f(\mathbf{y}) - f(\mathbf{x}) \leq D(\mathcal{A}) \|\mathbf{x} - \mathbf{y}\|_2,$$

so we obtain

$$|f(\mathbf{x}) - f(\mathbf{y})| \leq D(\mathcal{A}) \|\mathbf{x} - \mathbf{y}\|_2.$$

By Theorem 3.2.1 the following tail bound holds

$$P(|Z - E[Z]| \geq t) \leq 2 \exp\left(-\frac{t^2}{2D^2(\mathcal{A})}\right).$$

We define the quantity $\mathcal{G}(\mathcal{A}) := E[Z]$ as the *Gaussian complexity* of any subset $\mathcal{A}$ of $\mathbb{R}^n$.

The theory of sub-Gaussian variables can now be connected with the original motivation of further understanding how to link the structure of a function class $\mathcal{F}$ to the error bound.

¹If $\mathcal{A}$ is not closed we can choose an appropriate $\mathbf{a}^*$ in the closure of $\mathcal{A}$ s.t. $\|\mathbf{a}^*\|_2 = D(\mathcal{A})$

Chapter 4

Critical Inequality

In this chapter, we discuss critical inequalities, their solutions, and how they can be used in the context of concentration inequalities, to yield uniform bounds over an underlying function class $\mathcal{F}$. Furthermore, we define *star-shaped* function classes and how this property relates to critical inequalities.

4.1 From Gaussian complexity to uniform tail bounds

While the previous definitions deal with subsets of $\mathbb{R}^n$ we can also deduce useful tools into gain more insight to the size of a function class:

Let $\mathcal{H}$ be a class of real-valued functions with a common domain $\mathcal{X}$ and $\mathbf{x}_1^n := \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ are distinct points $\mathcal{X}$, then we define

$$\mathcal{H}(\mathbf{x}_1^n) := \{(f(\mathbf{x}_1), \dots, f(\mathbf{x}_n)) \in \mathbb{R}^n \mid f \in \mathcal{H}\}.$$

Since this is a subset of $\mathbb{R}^n$, we can calculate the Gaussian complexity with respect to the given function class $\mathcal{F}$. We will use a localized form of the Gaussian complexity:

Let $\mathcal{H}$ be a function class as above, then for a given radius $\delta > 0$ we define the *local Gaussian complexity* to be

$$\mathcal{G}_n(\delta; \mathcal{H}) := E \left[\sup_{\substack{g \in \mathcal{H} \\ \|g\|_n \leq \delta}} \left| \frac{1}{n} \sum_{i=1}^n w_i g(\mathbf{x}_i) \right| \right],$$

where $w_1, \dots, w_n$ are i.i.d. $\mathcal{N}(0, 1)$ random variables. In the setting of the standard non-parametric regression problem finding a radius $\delta > 0$ that satisfies the *critical inequality*

$$\frac{\mathcal{G}(\delta; \mathcal{H})}{\delta} \leq \frac{\delta}{2\sigma}, \quad (4.1.1)$$

where $\mathcal{H}$ is some function class, which will be essential to bounding the prediction error. The existence of such a solution depends on the structure of the

function class. A function class $\mathcal{H}$ is called star-shaped if for $g \in \mathcal{H}$ and $\alpha \in [0, 1]$ the function αg is also in $\mathcal{H}$. This property is sufficient for the existence of a solution to the critical inequality (4.1.1):

Theorem 4.1.1. If $\mathcal{H}$ is a star-shaped function class then the critical inequality (4.1.1) has a solution. More precisely the mapping $\nu \mapsto \frac{\mathcal{G}(\nu; \mathcal{H})}{\nu}$ is non-increasing, so any inequality of the form

$$\frac{\mathcal{G}(\delta; \mathcal{H})}{\delta} \leq c\delta \quad \text{for } c > 0$$

has a solution δ_n and all $u \geq \delta_n$ are also solutions.

Proof. Let $0 < \nu < \mu$ and $g \in \mathcal{H}$, such that $\|g\|_n \leq \mu$, then

$$\frac{|\frac{1}{n} \sum_{i=1}^n w_i g(\mathbf{x}_i)|}{\mu} = \frac{|\frac{1}{n} \sum_{i=1}^n w_i \frac{\nu}{\mu} g(\mathbf{x}_i)|}{\nu}.$$

The function $\tilde{g} := \nu g / \mu$ satisfies $\|\tilde{g}\|_n \leq \nu$ and $\tilde{g} \in \mathcal{H}$, because $\nu / \mu \in [0, 1]$, $g \in \mathcal{H}$ and $\mathcal{H}$ is star-shaped. Now applying suprema yields

$$\sup_{\substack{g \in \mathcal{H} \\ \|g\|_n \leq \mu}} \frac{|\frac{1}{n} \sum_{i=1}^n w_i g(\mathbf{x}_i)|}{\mu} = \sup_{\substack{g \in \mathcal{H} \\ \|g\|_n \leq \nu}} \frac{|\frac{1}{n} \sum_{i=1}^n w_i \frac{\nu}{\mu} g(\mathbf{x}_i)|}{\nu} \leq \sup_{\substack{g \in \mathcal{H} \\ \|g\|_n \leq \nu}} \frac{|\frac{1}{n} \sum_{i=1}^n w_i g(\mathbf{x}_i)|}{\nu}.$$

Applying expectations with respect to the distribution of the Gaussian vector $\mathbf{w} = (w_1, \dots, w_n)$ yields the claim, since the expectation of the term on the left-hand side is $\mathcal{G}(\mu; \mathcal{H})$ and the expectation of the term on the right-hand side is $\mathcal{G}(\nu; \mathcal{H})$. $\square$

Remark: The proof above makes no use of the fact that $w_1, \dots, w_n$ are Gaussian, so Theorem 4.1.1 also holds for arbitrary random variables $v_1, \dots, v_n$. We will use this in Chapter 6, when dealing with so called *Rademacher complexities* instead of Gaussian complexities.

Remember the motivation to find uniform bounds for terms of the form

$$\left| \sum_{i=1}^n w_i g(\mathbf{x}_i) \right|.$$

We now state a Lemma that will be key in proving bounds on the prediction error:

Lemma 4.1.1. Let $\mathcal{H}$ be a star-shaped function class, let $\delta_n > 0$ be a solution to the critical inequality (4.1.1) and define for any positive u

$$\mathcal{A}(u) := \left\{ \exists g \in \mathcal{H} \cap \{\|g\|_n \geq u\} \mid \frac{\sigma}{n} \left| \sum_{i=1}^n w_i g(\mathbf{x}_i) \right| \geq 2\|g\|_n u \right\},$$

then for all $u \geq \delta_n$ the following tail-bound holds

$$P(\mathcal{A}(u)) \leq e^{-\frac{nu^2}{2\sigma^2}}.$$

Remark: Note that the Lemma implies that for *any* fixed $g \in \mathcal{H}$ such that $\|g\|_n \geq u$ the event

$$\left\{ \frac{\sigma}{n} \left| \sum_{i=1}^n w_i g(\mathbf{x}_i) \right| \geq 2\|g\|_n u \right\}$$

is a subset of $\mathcal{A}(u)$, so if Lemma 4.1.1 holds we find the bound

$$P\left(\frac{\sigma}{n} \left| \sum_{i=1}^n w_i g(\mathbf{x}_i) \right| \geq 2\|g\|_n u\right) \leq P(\mathcal{A}(u)) \leq e^{-\frac{nu^2}{2\sigma^2}} \quad (4.1.2)$$

for all such g . In the original problem, we aim to control the quantity

$$\left| \sum_{i=1}^n w_i \tilde{\Delta}(\mathbf{x}_i) \right|,$$

so naturally we would apply the Lemma 4.1.1 above to the function class $\partial\mathcal{F} := \{f_1 - f_2 \mid f_1, f_2 \in \mathcal{F}\}$ or some appropriate subset thereof, since $\tilde{\Delta} \in \partial\mathcal{F}$. We prove Lemma 4.1.1:

Proof. Suppose there exists $g \in \mathcal{H}$ s.t. $\|g\|_n \geq u$ and

$$\frac{\sigma}{n} \left| \sum_{i=1}^n w_i g(\mathbf{x}_i) \right| \geq 2\|g\|_n u.$$

Since $\mathcal{H}$ is star-shaped $\tilde{g} := \frac{u}{\|g\|_n} g$ lies in $\mathcal{H}$ again and $\|\tilde{g}\|_n = u$. This yields

$$\frac{\sigma}{n} \left| \sum_{i=1}^n w_i \tilde{g}(\mathbf{x}_i) \right| = \frac{u}{\|g\|_n} \frac{\sigma}{n} \left| \sum_{i=1}^n w_i g(\mathbf{x}_i) \right| \geq 2u^2$$

and so $P(\mathcal{A}(u)) \leq P(Z_n(u) \geq 2u^2)$, where

$$Z_n(u) := \sup_{\substack{\tilde{g} \in \mathcal{H} \\ \|\tilde{g}\|_n \leq u}} \frac{\sigma}{n} \left| \sum_{i=1}^n w_i \tilde{g}(\mathbf{x}_i) \right|$$

If we treat $Z_n(u)$ as a function in $(w_1, \dots, w_n)$, then it is $\frac{\sigma u}{\sqrt{n}}$ -Lipschitz. This can be seen by applying the reverse triangle inequality and then proceeding exactly as in Example 3.2.1. By Theorem 3.2.1 $P(Z_n(u) \geq s + E[Z_n(u)]) \leq e^{-\frac{ns^2}{2\sigma^2 u^2}}$, where $s > 0$. Let $s = u^2$, we now want to prove that $2u^2 \geq u^2 + E[Z_n(u)]$ which would yield the claim: Note that $E[Z_n(u)] = \sigma \mathcal{G}_n(u; \mathcal{H})$, then

$$E[Z_n(u)] = \sigma \mathcal{G}_n(u; \mathcal{H}) \leq \frac{u^2}{2} \leq u^2,$$

where in the first inequality we use the previously discussed fact, that $u \geq \delta_n$ is also a solution to the critical inequality. Putting together the pieces yields:

$$P(\mathcal{A}(u)) \leq P(Z_n(u) \geq 2u^2) \leq P(Z_n(u) \geq u^2 + E[Z_n(u)]) \leq e^{-\frac{nu^2}{2\sigma^2}}.$$

□

Further remark: The general approach was to show that the event $\mathcal{A}(u)$ implies the event $Z \geq 2u^2$, where Z is a supremum indexed over the function class $\mathcal{F}$. The critical inequality allows us to bring $E[Z]$ into the inequality and utilize a concentration inequality. We denote this since this is a recurring use of critical inequalities.

By our prior discussion, we want to apply Lemma 4.1.1 to $\partial\mathcal{F}$, so $\partial\mathcal{F}$ needs to be star-shaped. A sufficient condition for $\partial\mathcal{F}$ to be star-shaped is for $\mathcal{F}$ to be convex. $\mathcal{F}$ is called convex if for $f, g \in \mathcal{F}$ and $\alpha \in [0, 1]$ the function $\alpha f + (1 - \alpha)g$ also lies in $\mathcal{F}$.

Theorem 4.1.2. If the function class $\mathcal{F}$ is convex, then $\partial\mathcal{F}$ is star-shaped.

Proof. Let $g \in \partial\mathcal{F}$, then by definition of $\partial\mathcal{F}$ it has the form $f_1 - f_2$ for $f_1, f_2 \in \mathcal{F}$. Let $\alpha \in [0, 1]$, then by convexity of $\mathcal{F}$ the function $f^{(\alpha)} := \alpha f_1 + (1 - \alpha)f_2$ is in $\mathcal{F}$. Rearranging yields

$$\alpha g = \alpha(f_1 - f_2) = f^{(\alpha)} - f_2 \in \partial\mathcal{F}.$$

Since α was chosen arbitrarily in $[0, 1]$ this yields the claim. □

Remark: This ensures that we can apply Theorem 4.1.1 and Lemma 4.1.1 to $\partial\mathcal{F}$ under the condition that $\mathcal{F}$ is convex.

Chapter 5

Bounding the Prediction Error

In this chapter, we state and prove the oracle inequality [20, Theorem 13.17] by applying the theory from Chapter 4. We then briefly discuss how these risk bounds can be applied in practice, meaning for a concrete function class $\mathcal{F}$.

5.1 Oracle inequality

We are now ready to state a bound on the prediction error $\|\hat{f} - f^*\|_n$. We do so without assuming that f^* is a member of the underlying function class $\mathcal{F}$. Error bounds for such problems include an *approximation term* of the form $\|f - f^*\|_n$ for $f \in \mathcal{F}$. This term can't be computed without prior knowledge of the vector $(f^*(\mathbf{x}_1), \dots, f^*(\mathbf{x}_n))$, hence we refer to inequalities of this type as *oracle inequalities*.

Remember the definition of $\hat{f}$:

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \left\{ \frac{1}{2n} \sum_{i=1}^n (y_i - f(\mathbf{x}_i))^2 + \lambda_n \|f\|_{\mathcal{F}}^2 \right\}$$

and the basic inequality (2.3.3)

$$\frac{1}{2} \|\hat{\Delta}\|_n^2 \leq \frac{1}{2} \|\tilde{f} - f^*\|_n^2 + \frac{\sigma}{n} \left| \sum_{i=1}^n w_i \tilde{\Delta}(\mathbf{x}_i) \right| + \lambda_n (\|\tilde{f}\|_{\mathcal{F}}^2 - \|\hat{f}\|_{\mathcal{F}}^2),$$

which was derived only by using the definition of $\hat{f}$ and the fact that $y_i = f^*(\mathbf{x}_i) + \sigma w_i$. In the specific instance of the theorem to be proven we will need the following form of localized Gaussian complexity with respect to function class $\mathcal{F}$:

$$\mathcal{G}_n(\delta, \mathbb{B}_{\partial \mathcal{F}}(3)) = E_{\mathbf{w}} \left[\sup_{\substack{g \in B_{\partial \mathcal{F}}(3), \\ \|g\|_n \leq \delta}} \left| \frac{1}{n} \sum_{i=1}^n w_i g(\mathbf{x}_i) \right| \right],$$

which we denote as $\mathcal{G}_n(\delta)$ here for notation purposes, where $\mathbb{B}_{\partial\mathcal{F}}(3) := \{f \in \partial\mathcal{F} \mid \|f\|_{\mathcal{F}} \leq 3\}$ and $E_{\mathbf{w}}$ denotes the expected value with respect to the Gaussian vector $\mathbf{w} = (w_1, \dots, w_n)$. For a user defined radius $R > 0$ let $\delta_n > 0$ be a solution to the inequality

$$\frac{\mathcal{G}_n(\delta)}{\delta} \leq \frac{R}{2\sigma}\delta, \quad (5.1.1)$$

note that if $\partial\mathcal{F}$ is star-shaped a solution δ_n exists due to Theorem 4.1.1.

Theorem 5.1.1. Given the previously described observation model and a convex function class $\mathcal{F}$, suppose that we solve the convex program (2.2.3) with some regularization parameter $\lambda_n \geq 2\delta_n^2$, where δ_n is a solution to (5.1.1). Then there are universal positive constants c_0, c_1, c_2, c_3 such that

$$\|\hat{f} - f^*\|_n^2 \leq c_0 \inf_{\|f\|_{\mathcal{F}} \leq R} \|f - f^*\|_n^2 + c_1 R^2 (\delta_n^2 + \lambda_n) \quad (5.1.2)$$

with probability greater than $1 - c_2 e^{-c_3 \frac{nR^2\delta_n^2}{\sigma^2}}$.

Remark: This provides non-asymptotic control over the prediction error. Furthermore the bound consists of the *approximation error*

$$c_0 \inf_{\|f\|_{\mathcal{F}} \leq R} \|f - f^*\|_n^2$$

and the *estimation error*

$$c_1 R^2 (\delta_n^2 + \lambda_n).$$

Note that δ_n is also a solution to the critical inequality (5.1.1) for $r \geq R$ instead of R . This allows for bounds on the expected prediction risk $E[\|\hat{f} - f^*\|_n^2]$ using tail integration, see Corollary 5.2.1.

5.1.1 Proof of Theorem 5.1.1 (Oracle inequality)

Proof. Given the observation model $y_i = f^*(\mathbf{x}_i) + \sigma w_i$, we can re-scale by R^{-1} and obtain an equivalent model with noise variance $(\frac{\sigma}{R})^2$. The inequality then re-scales ($\hat{f}$ and f^* now from the re-scaled model) to

$$\|\hat{f} - f^*\|_n^2 \leq c_0 \inf_{\|f\|_{\mathcal{F}} \leq 1} \|f - f^*\|_n^2 + c_1 (\delta_n^2 + \lambda_n).$$

We prove the claim for this re-scaled model and then simply need to re-scale by R^2 . For ease of notation we introduce $\tilde{\sigma} := \sigma/R$ and assume $\tilde{f} \in \partial\mathcal{F}$, s.t. $\|\tilde{f}\|_{\mathcal{F}} \leq 1$.

Recall the definitions $\hat{\Delta} := \hat{f} - f^*$ and $\tilde{\Delta} := \tilde{f} - \tilde{f}$ and that the given observation model $y_i = f^*(\mathbf{x}_i) + \tilde{\sigma} w_i$ yields the inequality

$$\frac{1}{2} \|\hat{\Delta}\|_n^2 \leq \frac{1}{2} \|\tilde{f} - f^*\|_n^2 + \frac{\sigma}{n} \left| \sum_{i=1}^n w_i \tilde{\Delta}(\mathbf{x}_i) \right| + \lambda_n (\|\tilde{f}\|_{\mathcal{F}}^2 - \|\hat{f}\|_{\mathcal{F}}^2). \quad (5.1.3)$$

Since $\|\tilde{f}\|_{\mathcal{F}} \leq 1$ this yields

$$\|\tilde{f}\|_{\mathcal{F}}^2 - \|\hat{f}\|_{\mathcal{F}}^2 \leq 1 - \|\hat{f}\|_{\mathcal{F}}^2 \leq 1$$

and inserting into (5.1.3) yields the possibly weaker bound

$$\frac{1}{2}\|\hat{\Delta}\|_n^2 \leq \frac{1}{2}\|\tilde{f} - f^*\|_n^2 + \frac{\sigma}{n} \left| \sum_{i=1}^n w_i \tilde{\Delta}(\mathbf{x}_i) \right| + \lambda_n. \quad (5.1.4)$$

We divide the proof into different cases and sub-cases and then prove that within these cases the claimed constants exist. These constants might differ in each case but we can combine the results by taking according minima and maxima of constants for the final result:

Case 1: If $\|\tilde{\Delta}\|_n \leq \delta_n$ we can yield the claim via direct arguments:

$$\begin{aligned} \|\hat{\Delta}\|_n^2 &= \|(\tilde{f} - f^*) + \tilde{\Delta}\|_n^2 \\ &\stackrel{i)}{\leq} (\|\tilde{f} - f^*\|_n + \delta_n)^2 \\ &= \|\tilde{f} - f^*\|_n^2 + 2\|\tilde{f} - f^*\|_n \delta_n + \delta_n^2 \\ &\stackrel{ii)}{\leq} (1 + 2\beta)\|\tilde{f} - f^*\|_n^2 + (1 + \frac{2}{\beta})\delta_n^2 \quad \forall \beta > 0, \end{aligned}$$

where we use the triangle inequality and $\|\tilde{\Delta}\|_n \leq \delta_n$ in *i*) and Cauchy inequality with epsilon to bound $\|\tilde{f} - f^*\|_n \delta_n$ in *ii*). Since this bound holds for all $\tilde{f}$ we can apply the infimum (and could even optimize for $\beta > 0$).

Case 2: We therefore assume $\|\tilde{\Delta}\|_n > \delta_n$ and divide into further sub-cases:

a) Suppose $\|\hat{f}\|_n \leq 2$. The triangle inequality and $\|\tilde{f}\|_n \leq 1$ yields $\|\tilde{\Delta}\|_n \leq 3$. Consequently we can apply Lemma 4.1.1 on $\{f \in \partial\mathcal{F} \mid \|f\|_{\mathcal{F}} \leq 3\}$, which yields:

$$P(\mathcal{A}(\delta_n)) \leq e^{-\frac{n\delta_n^2}{2\sigma^2}},$$

where

$$\mathcal{A}(\delta_n) := \left\{ \exists g \in \{f \in \partial\mathcal{F} \mid \|f\|_{\mathcal{F}} \leq 3\} \cap \{\|g\|_n \geq \delta_n\} \mid \frac{\sigma}{n} \left| \sum_{i=1}^n w_i g(\mathbf{x}_i) \right| \geq 2\|g\|_n \delta_n \right\}.$$

By equation (4.1.2) this implies

$$\frac{\tilde{\sigma}}{n} \left| \sum_{i=1}^n w_i \tilde{\Delta}(\mathbf{x}_i) \right| \leq 2\delta_n \|\tilde{\Delta}\|_n \quad \text{with probability at least } 1 - e^{-\frac{n\delta_n^2}{2\sigma^2}}. \quad (5.1.5)$$

Note that strictly speaking we are now treating the function class $\partial\mathcal{F}/R$ instead of $\partial\mathcal{F}$, so it is crucial that δ_n is still a solution to the critical inequality (5.1.1) for $\partial\mathcal{F}/R$ instead of $\partial\mathcal{F}$, so we can apply Lemma 4.1.1. It suffices to show $\mathcal{G}_n(\delta_n, \mathbb{B}_{\partial\mathcal{F}/R}(3)) \leq \mathcal{G}_n(\delta_n, \mathbb{B}_{\partial\mathcal{F}}(3))$, since then

$$\frac{\mathcal{G}_n(\delta_n, \mathbb{B}_{\partial\mathcal{F}/R}(3))}{\delta_n} \leq \frac{\mathcal{G}_n(\delta_n, \mathbb{B}_{\partial\mathcal{F}}(3))}{\delta_n} \leq \frac{\delta_n}{2\tilde{\sigma}},$$

where the second inequality holds, since δ_n is a solution to the critical inequality (5.1.1). We now show $\mathcal{G}_n(\delta_n, \mathbb{B}_{\partial\mathcal{F}/R}(3)) \leq \mathcal{G}_n(\delta, \mathbb{B}_{\partial\mathcal{F}}(3))$:

For $R \geq 1$ the function class $\partial\mathcal{F}/R$ is a subset of $\partial\mathcal{F}$, since $\partial\mathcal{F}$ is star-shaped and hence $\mathcal{G}_n(\delta_n, \mathbb{B}_{\partial\mathcal{F}/R}(3)) \leq \mathcal{G}_n(\delta, \mathbb{B}_{\partial\mathcal{F}}(3))$. For $0 < R < 1$ the following implications hold:

$$\|f/R\|_{\mathcal{F}} \leq 3, \|f/R\|_n \leq \delta \Leftrightarrow \|f\|_{\mathcal{F}} \leq 3R, \|f\|_n \leq \delta R \Rightarrow \|f\|_{\mathcal{F}} \leq 3, \|f\|_n \leq \delta,$$

which implies $\mathcal{G}_n(\delta_n, \mathbb{B}_{\partial\mathcal{F}/R}(3)) \leq \mathcal{G}_n(\delta, \mathbb{B}_{\partial\mathcal{F}}(3))$.

We return to (5.1.5) and further bound:

$$\begin{aligned} 2\delta_n \|\tilde{\Delta}\|_n &\stackrel{i)}{\leq} 2\delta_n \|\hat{\Delta}\|_n + 2\delta_n \|\tilde{f} - f^*\|_n \\ &\stackrel{ii)}{\leq} 2\delta_n \|\hat{\Delta}\|_n + 2\delta_n^2 + \frac{\|\tilde{f} - f^*\|_n^2}{2}, \end{aligned}$$

where we use the triangle inequality in $i)$ and the Cauchy inequality with epsilon in $ii)$. Inserting in to the weaker basic inequality (5.1.4) then yields:

$$\frac{1}{2} \|\hat{\Delta}\|_n^2 \leq \|\tilde{f} - f^*\|_n^2 + 2\delta_n \|\hat{\Delta}\|_n + 2\delta_n^2 + \lambda_n.$$

Using the quadratic formula yields the bound

$$\|\hat{\Delta}\|_n \leq 2\delta_n + \sqrt{6\delta_n^2 + \|\tilde{f} - f^*\|_n^2 + \lambda_n}.$$

Squaring yields

$$\begin{aligned} \|\hat{\Delta}\|_n^2 &\leq 10\delta_n^2 + \|\tilde{f} - f^*\|_n^2 + \lambda_n + 4\delta_n \sqrt{6\delta_n^2 + \|\tilde{f} - f^*\|_n^2 + \lambda_n} \\ &\leq 3\|\tilde{f} - f^*\|_n^2 + 22\delta_n^2 + 2\lambda_n, \end{aligned}$$

where we apply the Cauchy inequality $ab \leq \frac{1}{2}(a^2 + b^2)$ in the last inequality.

$b)$ Now assume $\|\hat{f}\|_{\mathcal{F}} > 2 > 1 \geq \|\tilde{f}\|_{\mathcal{F}}$. We now want to bound the term $\lambda_n(\|\tilde{f}\|_{\mathcal{F}}^2 - \|\hat{f}\|_{\mathcal{F}}^2)$ in (5.1.3):

$$\|\tilde{f}\|_{\mathcal{F}}^2 - \|\hat{f}\|_{\mathcal{F}}^2 = \underbrace{(\|\tilde{f}\|_{\mathcal{F}} + \|\hat{f}\|_{\mathcal{F}})}_{>1} \underbrace{(\|\tilde{f}\|_{\mathcal{F}} - \|\hat{f}\|_{\mathcal{F}})}_{<0} \leq \underbrace{\|\tilde{f}\|_{\mathcal{F}} - \|\hat{f}\|_{\mathcal{F}}}_{<0}.$$

Note that $\hat{f} = \tilde{f} + \tilde{\Delta}$ and by the triangle inequality $\|\hat{f}\|_{\mathcal{F}} \geq \|\tilde{\Delta}\|_{\mathcal{F}} - \|\tilde{f}\|_{\mathcal{F}}$, this yields

$$\begin{aligned} \lambda_n(\|\tilde{f}\|_{\mathcal{F}}^2 - \|\hat{f}\|_{\mathcal{F}}^2) &\leq \lambda_n(\|\tilde{f}\|_{\mathcal{F}} - \|\hat{f}\|_{\mathcal{F}}) \\ &\leq \lambda_n(2\|\tilde{f}\|_{\mathcal{F}} - \|\tilde{\Delta}\|_{\mathcal{F}}) \\ &\leq \lambda_n(2 - \|\tilde{\Delta}\|_{\mathcal{F}}), \end{aligned}$$

where we use $\|\hat{f}\|_{\mathcal{F}} \geq \|\tilde{\Delta}\|_{\mathcal{F}} - \|\tilde{f}\|_{\mathcal{F}}$ in the second inequality and the assumption $\|\tilde{f}\|_{\mathcal{F}} \leq 1$ in the last bound.

Substituting into the basic inequality (5.1.3) yields

$$\frac{1}{2}\|\hat{\Delta}_n\|^2 \leq \frac{1}{2}\|\tilde{f} - f^*\|_n^2 + \frac{\tilde{\sigma}}{n} \left| \sum_{i=1}^n w_i \tilde{\Delta}(\mathbf{x}_i) \right| + \lambda_n(2 - \|\tilde{\Delta}\|_{\mathcal{F}}). \quad (5.1.6)$$

Again the aim is to now control the stochastic component such that we can reduce the problem to applying the quadratic formula again. Since now $\|\tilde{\Delta}\|_{\mathcal{F}}$ may be larger than 3 we can not immediately apply Lemma 4.1.1 and a more refined approach is needed:

Lemma 5.1.1. There are universal constants (c_1, c_2) such that with probability greater than $1 - c_1 e^{-c_2 \frac{n\delta_n^2}{\sigma^2}}$

$$\frac{\tilde{\sigma}}{n} \left| \sum_{i=1}^n w_i \Delta(\mathbf{x}_i) \right| \leq 2\delta_n \|\Delta\|_n + 2\delta_n^2 \|\Delta\|_{\mathcal{F}} + \frac{1}{16} \|\Delta\|_n^2 \quad (5.1.7)$$

uniformly for all $\Delta \in \partial\mathcal{F}$ such that $\|\Delta\|_{\mathcal{F}} \geq 1$.

We complete the proof using Lemma 5.1.1: Since $\|\tilde{f}\|_{\mathcal{F}} \leq 1$ and $\|\hat{f}\|_{\mathcal{F}} > 2$ by triangle inequality $\|\tilde{\Delta}\|_{\mathcal{F}} \geq \|\hat{f}\|_{\mathcal{F}} - \|\tilde{f}\|_{\mathcal{F}} > 1$. We can therefore apply Lemma 5.1.1. Inserting into the derived inequality (5.1.7) yields

$$\begin{aligned} \frac{1}{2}\|\hat{\Delta}_n\|^2 &\leq \frac{1}{2}\|\tilde{f} - f^*\|_n^2 + 2\delta_n \|\tilde{\Delta}\|_n + (2\delta_n^2 - \lambda_n) \|\tilde{\Delta}\|_{\mathcal{F}} + \frac{1}{16} \|\tilde{\Delta}\|_n^2 + 2\lambda_n \\ &\leq \frac{1}{2}\|\tilde{f} - f^*\|_n^2 + 2\delta_n \|\tilde{\Delta}\|_n + \frac{1}{16} \|\tilde{\Delta}\|_n^2 + 2\lambda_n, \end{aligned} \quad (5.1.8)$$

where we use $2\delta_n^2 - \lambda_n \leq 0$ by assumption. By the triangle inequality $\|\tilde{\Delta}\|_n \leq \|\tilde{f} - f^*\|_n + \|\hat{\Delta}\|_n$ and similarly by the inequality $(a+b)^2 \leq 2a^2 + 2b^2$

$$\|\tilde{\Delta}\|_n^2 \leq 2\|\tilde{f} - f^*\|_n^2 + 2\|\hat{\Delta}\|_n^2.$$

Inserting this into (5.1.8) yields

$$\frac{3}{8}\|\hat{\Delta}_n\|^2 \leq \frac{5}{8}\|\tilde{f} - f^*\|_n^2 + 2\delta_n \|\tilde{f} - f^*\|_n + 2\delta_n \|\hat{\Delta}\|_n + 2\lambda_n,$$

applying the quadratic formula and the basic inequality $ab \leq \frac{1}{2}(a^2 + b^2)$, to control square-root terms and terms of the form $\delta_n \|\tilde{f} - f^*\|_n$ this yields

$$\|\tilde{\Delta}\|_n^2 \leq c_1 \|\tilde{f} - f^*\|_n^2 + c_2 \delta_n^2 + c_3 \lambda_n$$

for some positive constants c_1, c_2, c_3 , which yields the claim. $\square$

We prove Lemma 5.1.1:

Proof. Note that it is enough to prove the bound (5.1.7) for functions $g \in \partial\mathcal{F}$ such that $\|g\|_{\mathcal{F}} = 1$. Suppose it holds for such functions and we are given a function $\Delta \in \partial\mathcal{F}$, such that $\|\Delta\|_{\mathcal{F}} > 1$, then the claim holds for $\Delta/\|\Delta\|_{\mathcal{F}}$:

$$\frac{1}{\|\Delta\|_{\mathcal{F}}} \frac{\tilde{\sigma}}{n} \left| \sum_{i=1}^n w_i \Delta(\mathbf{x}_i) \right| \leq \frac{1}{\|\Delta\|_{\mathcal{F}}} \left(2\delta_n \|\Delta\|_n + 2\delta_n^2 \|\Delta\|_{\mathcal{F}} + \frac{1}{16} \frac{\|\Delta\|_n^2}{\|\Delta\|_{\mathcal{F}}} \right)$$

and hence

$$\begin{aligned} \frac{\tilde{\sigma}}{n} \left| \sum_{i=1}^n w_i \Delta(\mathbf{x}_i) \right| &\leq 2\delta_n \|\Delta\|_n + 2\delta_n^2 \|\Delta\|_{\mathcal{F}} + \frac{1}{16} \frac{\|\Delta\|_n^2}{\|\Delta\|_{\mathcal{F}}} \\ &\leq 2\delta_n \|\Delta\|_n + 2\delta_n^2 \|\Delta\|_{\mathcal{F}} + \frac{1}{16} \|\Delta\|_n^2, \end{aligned}$$

where the second inequality follows by $\|\Delta\|_{\mathcal{F}} > 1$.

Define $t_m := 2^m \delta_n$ for $m = 0, 1, \dots$. We will prove the bound (5.1.7) by decomposing with respect to $\|g\|_n$: For $0 \leq a < b$ denote $\mathcal{E}(a, b)$ as the event that the bound is violated for some $g \in \partial\mathcal{F}$ such that $\|g\|_{\mathcal{F}} = 1$ and $\|g\|_n \in [a, b]$. The event $\mathcal{E}$ that the bound (5.1.7) is violated is exactly $\mathcal{E}(0, t_0) \cup \bigcup_{m=0}^{\infty} \mathcal{E}(t_m, t_{m+1})$, we can therefore use the union bound

$$P(\mathcal{E}) \leq P(\mathcal{E}(0, t_0)) + \sum_{m=0}^{\infty} P(\mathcal{E}(t_m, t_{m+1})).$$

We now bound the right-hand side and start with the series:

Suppose the event $\mathcal{E}(t_m, t_{m+1})$ holds, then there exists a $g \in \partial\mathcal{F}$ s.t. $\|g\|_{\mathcal{F}} = 1$ and $\|g\|_n \in [t_m, t_{m+1}]$ such that

$$\begin{aligned} \frac{\tilde{\sigma}}{n} \left| \sum_{i=1}^n w_i g(\mathbf{x}_i) \right| &\geq 2\delta_n \|g\|_n + 2\delta_n^2 + \frac{1}{16} \|g\|_n^2 \\ &\geq 2\delta_n t_m + 2\delta_n^2 + \frac{1}{16} t_m^2 \\ &= \delta_n t_{m+1} + 2\delta_n^2 + \frac{1}{64} t_{m+1}^2, \end{aligned}$$

where the first inequality follows by assumption, the second inequality by $\|g\|_n \geq t_m$ and the last step by $t_{m+1} = 2t_m$. Defining for $t > 0$

$$Z_n(t) := \sup_{\substack{\|g\|_{\mathcal{F}} \leq 1 \\ \|g\|_n \leq t}} \left| \frac{\tilde{\sigma}}{n} \sum_{i=1}^n w_i g(\mathbf{x}_i) \right|,$$

then by the above discussion the event of $\mathcal{E}(t_m, t_{m+1})$ implies

$$\begin{aligned} Z_n(t_{m+1}) &\geq \delta_n t_{m+1} + 2\delta_n^2 + \frac{1}{64} t_{m+1}^2 \\ &\geq \delta_n t_{m+1} + \frac{1}{64} t_{m+1}^2. \end{aligned}$$

Note that by an argument analogous to the proof of Lemma 4.1.1 $Z_n(t)$ as a function $w \mapsto f(w)$ is $\frac{\tilde{\sigma}t}{\sqrt{n}}$ -Lipschitz and therefore the tail-bound

$$P(Z_n(t) \geq E[Z_n(t)] + u) \leq e^{-\frac{nu^2}{2\tilde{\sigma}^2 t^2}} \quad \forall u \geq 0 \quad (5.1.9)$$

holds. Note that for $t > \delta_n$

$$E[Z_n(t)] \leq \tilde{\sigma}\mathcal{G}_n(t) = t \frac{\tilde{\sigma}\mathcal{G}_n(t)}{t} \stackrel{i)}{\leq} t \frac{\tilde{\sigma}\mathcal{G}_n(\delta_n)}{\delta_n} \stackrel{ii)}{\leq} t\delta_n,$$

where in $i)$ we use the fact that $\nu \mapsto \mathcal{G}_n(\nu)/\nu$ is non-increasing and in $ii)$ that δ_n is a solution to the critical inequality. Setting $u = t_{m+1}^2/64$, $t = t_{m+1}$ and combining the results yields

$$\begin{aligned} P(\mathcal{E}(t_m, t_{m+1})) &\leq P(Z_n(t_{m+1}) \geq \delta_n t_{m+1} + t_{m+1}^2/64) \\ &\leq P(Z_n(t_{m+1}) \geq E[Z_n(t_{m+1})] + t_{m+1}^2/64) \\ &\leq e^{-c_2 \frac{nt_{m+1}^2}{\tilde{\sigma}^2}} = e^{-c_2 \frac{n\delta_n^2 2^{2m+2}}{\tilde{\sigma}^2}} \end{aligned}$$

for a universal constants $c_2 > 0$.

It remains to bound $P(\mathcal{E}(0, \delta_n))$ (remember $t_0 = \delta_n$). Suppose the event $\mathcal{E}(0, \delta_n)$ holds, then there exists a $g \in \partial\mathcal{F}$ s.t. $\|g\|_{\mathcal{F}} = 1$ and $\|g\|_n \leq \delta_n$ and

$$\begin{aligned} \frac{\tilde{\sigma}}{n} \left| \sum_{i=1}^n w_i g(\mathbf{x}_i) \right| &\geq 2\delta_n \|g\|_n + 2\delta_n^2 + \frac{1}{16} \|g\|_n^2 \\ &\geq 2\delta_n^2, \end{aligned}$$

which implies

$$Z_n(\delta_n) \geq 2\delta_n^2.$$

If now $E[Z_n(\delta_n)] \leq \delta_n^2$, then using the tail-bound from Theorem 3.2.1 for $u = \delta_n^2$ we obtain

$$P(\mathcal{E}(0, \delta_n)) \leq P(Z_n(\delta_n) \geq 2\delta_n^2) \leq P(Z_n(\delta_n) \geq E[Z_n(\delta_n)] + \delta_n^2) \leq e^{-\frac{n\delta_n^2}{2\tilde{\sigma}^2}}.$$

Now

$$E[Z_n(\delta_n)] \leq \tilde{\sigma}\mathcal{G}_n(\delta_n) \leq \delta_n^2/2 \leq \delta_n^2,$$

where we use the fact that δ_n is a solution to the critical inequality in the second inequality. Putting together the pieces yields

$$\begin{aligned} P(\mathcal{E}) &\leq P(\mathcal{E}(0, t_0)) + \sum_{m=0}^{\infty} P(\mathcal{E}(t_m, t_{m+1})) \\ &\leq e^{-\frac{n\delta_n^2}{2\tilde{\sigma}^2}} + \sum_{m=0}^{\infty} e^{-c_2 \frac{n\delta_n^2 2^{2m+2}}{\tilde{\sigma}^2}} \\ &\leq c_3 e^{-c_4 \frac{n\delta_n^2}{\tilde{\sigma}^2}} \end{aligned}$$

for positive universal constants $c_3, c_4 > 0$. One way to see this (last inequality): Define

$$q := e^{-c_2 \frac{n\delta_n^2}{\sigma^2}} \quad (\text{Note } 0 < q < 1).$$

This yields

$$\begin{aligned} e^{-\frac{n\delta_n^2}{2\sigma^2}} + \sum_{m=0}^{\infty} e^{-c_2 \frac{n\delta_n^2}{\sigma^2} 2^{2m+2}} &= e^{-\frac{n\delta_n^2}{2\sigma^2}} + \sum_{m=0}^{\infty} q^{2^{2m+2}} \\ &\leq e^{-\frac{n\delta_n^2}{2\sigma^2}} + \sum_{m=2}^{\infty} q^m \\ &= e^{-\frac{n\delta_n^2}{2\sigma^2}} + q^2 \frac{1}{1-q} \\ &= e^{-\frac{n\delta_n^2}{2\sigma^2}} + q^2 \frac{1+q}{1+q^2} \\ &\leq e^{-\frac{n\delta_n^2}{2\sigma^2}} + q^2 + q^3 \\ &\leq e^{-\frac{n\delta_n^2}{2\sigma^2}} + e^{-2c_2 \frac{n\delta_n^2}{\sigma^2}} + e^{-3c_2 \frac{n\delta_n^2}{\sigma^2}}. \end{aligned}$$

Taking appropriate minima of the constants yields the claim. $\square$

5.2 Bounding expected prediction error

We now aim to use Theorem 5.1.1 to bound $E[\|\hat{f} - f^*\|_n^2]$. Note that until now we have only discussed the existence of solutions to the critical inequality (4.1.1), we now give a lower bound on a solution δ_n under the condition that the constant function $f \equiv 1$ is a member of the underlying function class. We use this in the proof of Corollary 5.2.1 to bound $E[\|\hat{f} - f^*\|_n^2]$.

Lemma 5.2.1. Let $\mathcal{H}$ be a star-shaped function class that contains the constant function $f \equiv 1$ and let $\delta \in (0, 1]$ be a solution to the critical inequality (4.1.1), then

$$\frac{8}{\pi} \frac{\sigma^2}{n} \leq \delta^2.$$

Proof. Using the fact that δ is a solution to the critical inequality and rearranging said inequality (4.1.1) yields

$$E \left[\sup_{\substack{g \in \mathcal{H} \\ \|g\|_n \leq \delta}} \left| \frac{1}{n} \sum_{i=1}^n w_i g(x_i) \right| \right] = \mathcal{G}_n(\delta; \mathcal{H}) \leq \frac{\delta^2}{2\sigma}.$$

Since $\mathcal{H}$ is star-shaped and the constant function $f \equiv 1$ lies in $\mathcal{H}$, the re-scaled function δf also lies in $\mathcal{H}$, since $\delta \in (0, 1]$. Furthermore, by linearity of norm

one has $\|\delta f\|_n = \delta$. Combining this yields

$$E\left[\left|\frac{1}{n} \sum_{i=1}^n \delta w_i\right|\right] = \frac{\delta}{\sqrt{n}} E\left[\left|\frac{1}{\sqrt{n}} \sum_{i=1}^n w_i\right|\right] \leq \mathcal{G}_n(\delta; \mathcal{H}) \leq \frac{\delta^2}{2\sigma}.$$

We can calculate the expectation on the left-hand side: Define

$$X := \frac{1}{\sqrt{n}} \sum_{i=1}^n w_i.$$

By properties of the normal distribution, this yields $X \sim \mathcal{N}(0, 1)$. By properties of the *folded normal distribution* $|X|$ this yields $E[|X|] = \sqrt{2/\pi}$, see [9]. Putting together the pieces yields

$$\frac{\delta}{\sqrt{n}} E[|X|] = \sqrt{\frac{2}{\pi}} \frac{\delta}{\sqrt{n}} \leq \frac{\delta^2}{2\sigma} \Leftrightarrow \frac{8}{\pi} \frac{\sigma^2}{n} \leq \delta^2.$$

□

We now state a bound for the expected squared prediction error $E[\|\hat{f} - f^*\|_n^2]$:

Corollary 5.2.1. Under the conditions of Theorem 5.1.1, additionally, the constant function $f \equiv 1$ is a member of $\partial\mathcal{F}$ and n is large enough such that $8\sigma^2/(\pi R^2 n) \leq 1$, then there exist universal positive constants c'_0, c'_1 such that

$$E[\|\hat{f} - f^*\|_n^2] \leq c'_0 \inf_{\|f\|_{\mathcal{F}} \leq R} \|f - f^*\|_n^2 + c'_1 R^2 (\delta_n^2 + \lambda_n). \quad (5.2.1)$$

Proof. The aim is to use the layer cake formula (see [3, p. 273-276])

$$E[\|\hat{f} - f^*\|_n^2] = \int_0^\infty P(\|\hat{f} - f^*\|_n^2 > t) dt.$$

Note that since δ_n is a solution to the critical inequality

$$\frac{\mathcal{G}_n(\delta)}{\delta} \leq \frac{R}{2\sigma},$$

it is also a solution to the critical inequality

$$\frac{\mathcal{G}_n(\delta)}{\delta} \leq \frac{r}{2\sigma} \delta \quad \forall r \geq R$$

and hence we can apply Theorem 5.1.1: For $r \geq R$ the bound

$$\|\hat{f} - f^*\|_n^2 \leq c_0 \inf_{\|f\|_{\mathcal{F}} \leq r} \|f - f^*\|_n^2 + c_1 r^2 (\delta_n^2 + \lambda_n)$$

with probability greater than $1 - c_2 e^{-c_3 \frac{nr^2 \delta_n^2}{\sigma^2}}$. Note that

$$\inf_{\|f\|_{\mathcal{F}} \leq r} \|f - f^*\|_n^2 \leq \inf_{\|f\|_{\mathcal{F}} \leq R} \|f - f^*\|_n^2$$

holds, since $R \leq r$. So the possibly weaker bound

$$\|\hat{f} - f^*\|_n^2 \leq c_0 \inf_{\|f\|_{\mathcal{F}} \leq R} \|f - f^*\|_n^2 + c_1 r^2 (\delta_n^2 + \lambda_n)$$

holds with probability greater than $1 - c_2 e^{-c_3 \frac{nr^2 \delta_n^2}{\sigma^2}}$. We define the function $b : [R, \infty) \rightarrow [0, \infty)$

$$b(r) := c_0 \inf_{\|f\|_{\mathcal{F}} \leq R} \|f - f^*\|_n^2 + c_1 r^2 (\delta_n^2 + \lambda_n)$$

and can now apply the layer cake formula

$$\begin{aligned} E[\|\hat{f} - f^*\|_n^2] &= \int_0^\infty P(\|\hat{f} - f^*\|_n^2 > t) dt \\ &= \underbrace{\int_0^{b(R)} P(\|\hat{f} - f^*\|_n^2 > t) dt}_{\leq b(R)} + \int_{b(R)}^\infty P(\|\hat{f} - f^*\|_n^2 > t) dt \end{aligned}$$

Since $b(R)$ is already of the claimed form in (5.2.1), it remains to bound the integral on the right-hand side:

$$\begin{aligned} \int_{b(R)}^\infty P(\|\hat{f} - f^*\|_n^2 > t) dt &\stackrel{i)}{=} c_1 (\delta_n^2 + \lambda_n) \int_R^\infty P(\|\hat{f} - f^*\|_n^2 > b(r)) 2r dr \\ &\stackrel{ii)}{\leq} c_1 c_2 (\delta_n^2 + \lambda_n) \int_R^\infty e^{-c_3 \frac{nr^2 \delta_n^2}{\sigma^2}} 2r dr \\ &= c_1 c_2 (\delta_n^2 + \lambda_n) \frac{\sigma^2}{c_3 n \delta_n^2} \int_R^\infty e^{-c_3 \frac{nr^2 \delta_n^2}{\sigma^2}} 2r \frac{c_3 n \delta_n^2}{\sigma^2} dr \\ &= \frac{c_1 c_2}{c_3} \frac{\sigma^2}{n} \frac{(\delta_n^2 + \lambda_n)}{\delta_n^2} \int_{\frac{c_3 n R^2 \delta_n^2}{\sigma^2}}^\infty e^{-u} du \\ &= e^{-\frac{n R^2 \delta_n^2}{\sigma^2}} \frac{c_1 c_2}{c_3} \frac{\sigma^2}{n} \frac{(\delta_n^2 + \lambda_n)}{\delta_n^2} \\ &\leq \frac{c_1 c_2}{c_3} \frac{\sigma^2}{n} \frac{(\delta_n^2 + \lambda_n)}{\delta_n^2} = c' \frac{\sigma^2}{n} \frac{(\delta_n^2 + \lambda_n)}{\delta_n^2} \quad \text{for } c' := \frac{c_1 c_2}{c_3}, \end{aligned}$$

where we use the substitution $b(r) = t$ in $i)$ and Theorem 5.1.1 in $ii)$. Recall the notation $\tilde{\sigma} = \sigma/R$, then δ_n is a solution to the critical inequality

$$\frac{\mathcal{G}_n(\delta)}{\delta} \leq \frac{\delta}{2\tilde{\sigma}}$$

and $\partial\mathcal{F}$ contains the constant function $f \equiv 1$. If $\delta_n \in (0, 1]$, then by Lemma 5.2.1 there exists a positive universal constant $c > 0$ such that

$$c \frac{\tilde{\sigma}^2}{n} \leq \delta_n^2 \Leftrightarrow \frac{\sigma^2}{n} \leq c^{-1} R^2 \delta_n^2.$$

In the case that $\delta_n > 1$, then by assumption one has

$$\frac{8}{\pi} \frac{\sigma^2}{R^2 n} \leq 1 < \delta_n^2 \Leftrightarrow \frac{\sigma^2}{n} \leq \frac{\pi}{8} R^2 \delta_n^2.$$

In both cases, there exists a positive universal constant $c > 0$ s.t.

$$\frac{\sigma^2}{n} \leq c^{-1} R^2 \delta_n^2.$$

Putting together the pieces yields

$$\int_{b(R)}^{\infty} P(\|\hat{f} - f^*\|_n^2 > t) dt \leq c' \frac{\sigma^2}{n} \frac{(\delta_n^2 + \lambda_n)}{\delta_n^2} \leq c^{-1} c' R^2 (\delta_n^2 + \lambda_n),$$

which yields the claim. $\square$

Remark: Note that we do not assume $f^* \in \mathcal{F}$. If we assume that $f^* \in \mathcal{F}$, one can formulate a tighter bound without the approximation error using the same techniques:

Define $\mathcal{F}^* := \{f - f^* \mid f \in \mathcal{F}\}$ and for $r \geq 0$ define $B_{\mathcal{F}^*}(r) := \{g \in \mathcal{F}^* \mid \|g\|_{\mathcal{F}} \leq r\}$

Corollary 5.2.2. Let $\mathcal{F}$ be star-shaped around f^* , meaning $\mathcal{F}^*$ is star-shaped, and let δ_n be a solution to the critical inequality

$$\frac{\mathcal{G}_n(\delta, B_{\mathcal{F}^*}(3))}{\delta} \leq \frac{R}{2\sigma} \delta$$

universal constants c_0, c_2, c_3 such that

$$\|\hat{f} - f^*\|_n^2 \leq c_0 R^2 (\delta_n^2 + \lambda_n) \quad (5.2.2)$$

with probability at least $1 - c_1 e^{-c_2 \frac{n R^2 \delta_n^2}{\sigma^2}}$. If additionally, the constant function $f \equiv 1$ is a member of $\mathcal{F}^*$ and n is large enough such that $8\sigma^2/(\pi R^2 n) \leq 1$, there exists a universal positive constant c'_0 such that

$$E[\|\hat{f} - f^*\|_n^2] \leq c'_0 R^2 (\delta_n^2 + \lambda_n)$$

Proof. See Appendix A. $\square$

5.3 Determining rates via metric entropy

In the previous sections bounds on the prediction error and the expected prediction error were derived in terms of the solution to a critical inequality that involves the localized Gaussian complexity. While the existence of solutions to the critical inequality was discussed it remains to present a methodology by which solutions can be calculated. Before giving a concrete method, some definitions are necessary:

Definition 5.3.1. For a bounded subset $\mathbb{T} \subset \mathbb{R}^d$, we call a set $\{\theta^1, \dots, \theta^n\}$ a δ -cover wrt. to the metric d , if for any $\theta \in \mathbb{T}$ there exists an index $i \in \{1, \dots, n\}$ such that $d(\theta, \theta_i) \leq \delta$. The δ -covering number $N(\delta; \mathbb{T}, d)$ is the cardinality of the smallest δ -cover.

Note that in the localized Gaussian complexity we are dealing with the following bounded subset of $\mathbb{R}^n$

$$B_n(\delta, \mathcal{H}) := \{(f(\mathbf{x}_1), \dots, f(\mathbf{x}_n)) \in \mathbb{R}^n \mid f \in \mathcal{H}, \|f\|_n \leq \delta\}.$$

Using a chaining argument it can be shown that any δ that satisfies the inequality

$$\frac{16}{\sqrt{n}} \int_0^\delta \sqrt{\log N(t; B_n(\delta, \mathcal{H}), \|\cdot\|_n)} dt \leq \frac{\delta^2}{4\sigma} \quad (5.3.1)$$

also satisfies the critical inequality

$$\frac{\mathcal{G}(\delta; \mathcal{H})}{\delta} \leq \frac{\delta}{2\sigma},$$

see [20, Cor. 13.7] for a proof. The quantity $\log N(\delta; \mathbb{T}, d)$ is called *metric entropy* of the set $\mathbb{T}$ wrt. d . Calculating the metric entropy of $B_n(\delta, \mathcal{H})$, where $\mathcal{H} = \partial\mathcal{F}$ or $\mathcal{H} = \mathcal{F}^*$ can therefore yield concrete rates by calculating solutions δ_n using 5.3.1 and then employing Theorem 5.1.1 or Corollary 5.2.2. We exemplify the method described above for the case of linear regression:

Example 5.3.1. We employ Corollary 5.2.2 and use the function class

$$\mathcal{F}_{\text{lin}} = \{f_\theta(\cdot) = \langle \cdot, \theta \rangle \mid \theta \in \mathbb{R}^d\}.$$

For simplicity we drop the constraint on $B_{\partial\mathcal{F}}(3)$ from Theorem 5.1.1 and set $R = 1$, the arguments to follow can be extended in a straightforward manner. Note that $\partial\mathcal{F}_{\text{lin}} = \mathcal{F}_{\text{lin}}$. Furthermore, it can be shown [20, ch. 5], that

$$\log N(t; B_n(\delta, \mathcal{F}_{\text{lin}})) \leq r \log\left(1 + \frac{2\delta}{t}\right),$$

where $r := \text{rank}(\mathbf{X})$ and $\mathbf{X} \in \mathbb{R}^{n \times d}$ is the *design matrix*, that contains $\mathbf{x}_i$ as its i -th row. We can now yield a solution to the critical inequality and insert

into (5.3.1):

$$\begin{aligned}
\frac{16}{\sqrt{n}} \int_0^\delta \sqrt{\log N(t; B_n(\delta, \mathcal{F}_{\text{lin}}), \|\cdot\|_n)} dt &\leq 16 \sqrt{\frac{r}{n}} \int_0^\delta \sqrt{\log\left(1 + \frac{2\delta}{t}\right)} du \\
&\stackrel{i)}{=} 16\delta \sqrt{\frac{r}{n}} \int_0^1 \sqrt{\log\left(1 + \frac{2}{u}\right)} dt \\
&\stackrel{ii)}{=} c\delta \sqrt{\frac{r}{n}},
\end{aligned}$$

where in *i*) the substitution t/δ and in *ii*) the fact that the integral is a data-independent constant, so c is a positive universal constant. Now solving the inequality

$$c\delta \sqrt{\frac{r}{n}} \leq \frac{\delta^2}{4\sigma}$$

yields

$$\sigma^2 \frac{r}{n} \leq c' \delta^2,$$

where c' is a positive universal constant. So in particular

$$(c')^{-1} \sigma_n^2 \frac{r}{n} = \delta_n^2$$

yields a solution δ_n for the critical inequality (5.1.1). Using Corollary 5.2.2 this implies that there exist universal positive constants c_0, c_1 s.t.

$$E[\|\hat{f} - f^*\|_n^2] \leq c_0 \inf_{\|f\|_{\mathcal{F}} \leq 1} \|f - f^*\|_n^2 + c_1 \left(\sigma^2 \frac{r}{n} + \lambda_n \right).$$

Note that we can invoke the same argument under the assumption that $f^* \in \mathcal{F}_{\text{lin}}$, since $\mathcal{F}_{\text{lin}}^* = \mathcal{F}_{\text{lin}} - f^* = \mathcal{F}_{\text{lin}}$ and using Corollary 5.2.2 yields

$$E[\|\hat{f} - f^*\|_n^2] \leq c_1 \left(\sigma^2 \frac{r}{n} + \lambda_n \right).$$

For non-parametric examples see [20, Ex. 13.10-13.11 & 13.19-13.21].

Chapter 6

Uniform Laws

In this chapter, we change the setting to variable covariates and explore how uniform laws, namely concentration inequalities that hold uniformly over a function class, can be used to deduce bounds on the excess risk. We then briefly discuss how the theoretical results can be used to derive excess risk bounds for concrete choices of the function class $\mathcal{F}$.

6.1 From fixed to variable covariates

We return to discuss the general problem set-up from Chapter 1. In Theorem 5.1.1 we derived bounds on the squared prediction error

$$\|\hat{f} - f^*\|_n^2 = \frac{1}{n} \sum_{i=1}^n (\hat{f}(\mathbf{x}_i) - f^*(\mathbf{x}_i))^2,$$

where we viewed the covariates $\{\mathbf{x}_i\}_{i=1}^n$ as fixed and $\hat{f}$ is a solution to the problem

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \left\{ \frac{1}{2n} \sum_{i=1}^n (y_i - f(\mathbf{x}_i))^2 + \lambda_n \|f\|_{\mathcal{F}}^2 \right\}. \quad (6.1.1)$$

We now want to move to bounds on the true risk

$$\|\hat{f} - f^*\|_2^2 = \int (\hat{f}(\mathbf{x}) - f^*(\mathbf{x}))^2 P(d\mathbf{x}), \quad (6.1.2)$$

where P is the probability distribution over the covariates X .

Note that for a fixed function $f \in L^2(P)$ the following connection between the empirical measure P_n and the true distribution holds, since the covariates $\{\mathbf{x}_i\}_{i=1}^n$ are sampled according to P :

$$E_X [\|f\|_n^2] = \|f\|_2^2. \quad (6.1.3)$$

Since the covariates $\{\mathbf{x}_i\}_{i=1}^n$ are sampled independently, the law of large numbers applies to the random variable $f(X)$ under mild conditions [3, ch. 22]. It is also

possible to derive a straightforward non-asymptotic result assuming that the fourth-moment of $f(X)$ exists: $E[f^4(X)] < \infty$. Then by a minor extension of the theory covered in Chapter 3 (see [20, ch.2]) we obtain the lower tail-bound

$$P(\|f\|_n^2 \geq \|f\|_2^2 - t) \leq e^{-\frac{nt^2}{2E[f^4(X)]}} \quad \text{for } t \geq 0.$$

If it is now possible to bound the fourth-moment by a multiple of the second moment we could obtain non-trivial lower tail bounds. We will present a result that extends this sort of bound to hold uniformly over a given function class $\mathcal{F}$ by imposing a uniform fourth-moment condition. While the result is interesting in its own right the aim is also to apply the result to bound the excess risk (6.1.2).

In the following sections the underlying random variable with respect to which expected values and probabilities are calculated is not always clear as opposed to Chapter 5, since there the only random variable was the Gaussian vector $\mathbf{w} = (w_1, \dots, w_n)$. In the following section, we denote symbols E, P with a lowercase symbol: E_Z, P_Z to denote expected values and probabilities with respect to the distribution of the random variable Z , where Z can be any (vector-valued or real-valued) random variable. We denote $E_{Z,Y}, P_{Z,Y}$ for joint expectations/distributions of (Z, Y) .

6.2 A one-sided uniform law

We begin by stating a one-sided uniform law that controls the random variable $\|f\|_2^2 - \|f\|_n^2/2$. To do so, we will have to define some necessary quantities. Similarly to Theorem 5.1.1 a form of complexity is necessary that measures the size of underlying function classes.

A *Rademacher variable* ε is a random variable that takes values in $\{-1, 1\}$ equiprobably. Let $\mathcal{A} \subset \mathbb{R}^n$, then we define the random variable

$$Z(\mathcal{A}) := \sup_{\mathbf{a} \in \mathcal{A}} \sum_{i=1}^n \varepsilon_i a_i,$$

where $\mathbf{a} = (a_1, \dots, a_n)$ and $\varepsilon_1, \dots, \varepsilon_n$ are i.i.d. Rademacher variables. The quantity $\mathcal{R}(\mathcal{A}) := E_{\varepsilon}[Z(\mathcal{A})]$ is called *Rademacher complexity* of the set $\mathcal{A}$, where $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)$. As with the result in Chapter 5 we will formulate bounds in terms of a localized¹ complexity measure:

For a given $\delta > 0$ and a function class $\mathcal{F}$, the *localized population Rademacher complexity* is defined as

$$\overline{\mathcal{R}}_n(\delta, \mathcal{F}) := E_{\mathbf{x}, \varepsilon} \left[\sup_{\substack{f \in \mathcal{F}, \\ \|f\|_n \leq \delta}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(\mathbf{x}_i) \right], \quad (6.2.1)$$

where $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_n)$ is a vector of i.i.d. samples from the underlying distribution P , and $\{\varepsilon_i\}_{i=1}^n$ are i.i.d. Rademacher variables that are independent of

¹The term “localized” refers to the complexity measure only depending on the behavior of the function class in δ -ball in the $\|\cdot\|_n$ -norm

the sequence $\{\mathbf{x}_i\}_{i=1}^n$. Furthermore, we require the function class $\mathcal{F}$ to satisfy the following *fourth-moment condition*: There exists a constant $C > 0$ such that

$$E_X[f^4(X)] \leq C^2 E[f^2(X)] \quad \forall f \in \mathcal{F} \text{ s.t. } \|f\|_2 \leq 1. \quad (6.2.2)$$

One could insist on the stricter global fourth-moment condition

$$E_X[f^4(X)] \leq C^2 E[f^2(X)] \quad \forall f \in \mathcal{F},$$

which obviously implies (6.2.2) and formulate the uniform law accordingly. However, in some instances, for example if $f(X)$ is zero-mean Gaussian, there exists a constant $C > 0$ such that

$$E[f^4(X)] \leq C^2 (E[f^2(X)])^2 \quad \forall f \in \mathcal{F},$$

which does not necessarily imply the global fourth-moment condition but does imply (6.2.2). Now again as in Theorem 5.1.1, we consider a critical inequality: Let the function class $\mathcal{F}$ satisfy the fourth-moment condition 6.2.2 with constant $C > 0$ then, consider the critical inequality

$$\frac{\overline{\mathcal{R}}_n(\delta, \mathcal{F})}{\delta} \leq \frac{\delta}{128C}, \quad (6.2.3)$$

where the constant C is from the fourth-moment condition (6.2.2). Note that a solution δ_n exists if $\mathcal{F}$ is star-shaped since the left-hand side is non-increasing as a function in δ , the argument is the same as in the proof of Theorem 4.1.1. With these conditions defined and the notation $B_2(\delta) := \{f \in \mathcal{F} \mid \|f\|_2 \leq \delta\}$ we are ready to state the one-sided uniform law:

Theorem 6.2.1 (Theorem 14.12 in [20]). Let $\mathcal{F}$ be a star-shaped function class such that the fourth-moment condition (6.2.2) holds and let n be large enough such that there exists a solution δ_n to the critical inequality (6.2.3), such that $\delta_n \leq 1$. Then there exist positive universal constants c_1, c_2 such that for all $\delta \in [\delta_n, 1]$

$$\|f\|_n^2 \geq \frac{1}{2} \|f\|_2^2 \quad \forall f \in \mathcal{F} \setminus B_2(\delta), \quad (6.2.4)$$

with probability at least $1 - c_1 e^{-c_2 \frac{n\delta^2}{C^2}}$.

Remarks:

- One can extend this Theorem from the function class $\mathcal{F} \setminus B_2(\delta)$ to all of $\mathcal{F}$ if the function class is *cone-shaped*, meaning for $f \in \mathcal{F}$ and $\alpha > 1$ the function αf is also a member of $\mathcal{F}$.
- Note that taking the complement of the event (6.2.4) implies that for *any* fixed $f \in \mathcal{F}$ and $\delta \in [\delta_n, 1]$

$$\|f\|_n^2 + \delta^2 \leq \frac{1}{2} \|f\|_2^2 \quad \text{with probability at most } c_1 e^{-c_2 \frac{n\delta^2}{C^2}}. \quad (6.2.5)$$

This can be seen as follows: If the function f lies in $B_2(\delta)$, then

$$\frac{1}{2}\|f\|_2^2 \leq \frac{1}{2}\delta^2,$$

which contradicts (6.2.5). If $\|f\|_2^2 > \delta^2$, then taking the complement of the event (6.2.4) yields

$$\|f\|_n^2 \leq \frac{1}{2}\|f\|_2^2 \text{ with probability at most } c_1 e^{-c_2 \frac{n\delta^2}{C^2}},$$

so the stronger bound (6.2.5) holds with probability at most $c_1 e^{-c_2 \frac{n\delta^2}{C^2}}$.

Before giving the proof, we show how Theorem 6.2.1 and Theorem 5.2.2 can be applied to yield bounds on the excess risk. The approach essentially follows the proof of [20, Corollary 14.15], where a proof is given for the cases where $\hat{f}$ is the least-squares estimate.

6.3 From prediction error to excess risk bounds

In Theorem 5.1.1 it is not required for f^* to be a member of $\mathcal{F}$, but this will be necessary to apply Theorem 6.2.1 directly, since we want to insert the function class $\mathcal{F}^*$ for $\mathcal{F}$ and therefore $\mathcal{F}^*$ has to be star-shaped. This implies $0 \in \mathcal{F}^*$, which is equivalent to $f^* \in \mathcal{F}$.

Let δ_n and ε_n be solutions to

$$\frac{\overline{\mathcal{R}}_n(\delta, \mathcal{F}^*)}{\delta} \stackrel{i)}{\leq} \frac{\delta}{128C} \quad \text{and} \quad \frac{\mathcal{G}_n(\varepsilon, \mathcal{F}^*)}{\varepsilon} \stackrel{ii)}{\leq} \frac{R\varepsilon}{2\sigma}$$

respectively. Note that ε_n is a random variable here, depending on the covariates $\{\mathbf{x}_i\}_{i=1}^n$.

Corollary 6.3.1. Under the conditions of Corollary 5.2.2 and Theorem 6.2.1 and furthermore $f^* \in \mathcal{F}$, there exist positive universal constants c_0, c_1, c_2, c_3 such that

$$\|\hat{f} - f^*\|_2^2 > c_0 R^2 (\delta_n^2 + \varepsilon_n^2 + \lambda_n) + c_1 \delta_n^2 \quad (6.3.1)$$

with probability at most $c_2 e^{-c_3 \frac{\min\{1, R^2\} n \delta_n^2}{\sigma^2 + C^2}}$.

Remark: The bound provides non-asymptotic control over the squared prediction error for the regularized least squares estimate. The result shows how utilizing Theorem 6.2.1 yields additional terms $R^2 \delta_n^2 + \delta_n^2$ when moving from the prediction error bound in Corollary 5.2.2 to an excess risk bound.

Proof. The main proof idea is to utilize (6.2.5) for $\hat{f} - f^*$ instead of f and $\mathcal{F}^*$ instead of $\mathcal{F}$. We then have to insert the bound from Corollary 5.2.2 appropriately to bound $\|\hat{f} - f^*\|_n^2$. We divide the proof into two cases:

Case 1: Let $\delta_n \geq \varepsilon_n$. This guarantees that δ_n is also a solution to *ii*), therefore we can apply Corollary 5.2.2 and this yields (constants taken from Corollary 5.2.2)

$$P_{\mathbf{w}}(\|\hat{f} - f^*\|_n^2 > c_0 R^2(\delta_n^2 + \lambda_n)) \leq c_1 e^{-c_2 \frac{n R^2 \delta_n^2}{\sigma^2}},$$

where $\mathbf{w}$ denotes the Gaussian noise-vector from the observation model (2.2.1). On the other hand, applying Theorem 6.2.1 (with renamed constants) yields

$$P_{\mathbf{x}, \mathbf{w}}(\|\hat{f} - f^*\|_2^2 \geq 2\delta_n^2 + 2\|\hat{f} - f^*\|_n^2) \leq c_3 e^{-c_4 \frac{n \delta_n^2}{C^2}}.$$

Combining these two pieces and using the fact that $\mathbf{x}$ and $\mathbf{w}$ are independent yields

$$P_{\mathbf{x}, \mathbf{w}}(\|\hat{f} - f^*\|_2^2 \geq 2\delta_n^2 + 2(c_0 R^2(\delta_n^2 + \lambda_n))) \leq c'_0 e^{-c'_1 \frac{\min\{1, R^2\} n \delta_n^2}{\sigma^2 + C^2}},$$

for universal positive constants c'_0, c'_1 .

Case 2: Let $\delta_n < \varepsilon_n$. We define the events

$$\begin{aligned} \mathcal{A} &:= \{\delta_n < \varepsilon_n\}, \quad \mathcal{B} := \{\|\hat{f} - f^*\|_n^2 \leq c_0 R^2(\varepsilon_n^2 + \lambda_n)\}, \\ \mathcal{E} &:= \{\|\hat{f} - f^*\|_2^2 > 2\delta_n^2 + 2(c_0 R^2(\varepsilon_n^2 + \lambda_n))\}. \end{aligned}$$

It suffices to bound the event $\mathcal{A} \cap \mathcal{E}$, we do so using the event $\mathcal{B}$ and monotonicity and additivity of the probability measure as follows

$$P_{\mathbf{x}, \mathbf{w}}(\mathcal{A} \cap \mathcal{E}) = P_{\mathbf{x}, \mathbf{w}}(\mathcal{A} \cap \mathcal{E} \cap \mathcal{B}) + P_{\mathbf{x}, \mathbf{w}}(\mathcal{A} \cap \mathcal{E} \cap \mathcal{B}^c) \leq P_{\mathbf{x}, \mathbf{w}}(\mathcal{E} \cap \mathcal{B}) + P_{\mathbf{x}, \mathbf{w}}(\mathcal{A} \cap \mathcal{B}^c).$$

On the one hand

$$P_{\mathbf{x}, \mathbf{w}}(\mathcal{E} \cap \mathcal{B}) \leq P_{\mathbf{x}, \mathbf{w}}(\|\hat{f} - f^*\|_2^2 \geq 2\delta_n^2 + 2\|\hat{f} - f^*\|_n^2) \leq c_3 e^{-c_4 \frac{n \delta_n^2}{C^2}}.$$

On the other hand, if we define $I_{\mathcal{A}}$ to be the indicator function of $\mathcal{A}$ and note that that this function is dependent on $\mathbf{x}$ and not on $\mathbf{w}$, since ε_n is only dependent on $\mathbf{x}$ not on $\mathbf{w}$, then using independence of $\mathbf{x}$ and $\mathbf{w}$ and Corollary 5.2.2 yields

$$P_{\mathbf{x}, \mathbf{w}}(\mathcal{A} \cap \mathcal{B}^c) = E_{\mathbf{x}}[P_{\mathbf{w}}(\mathcal{B}^c) I_{\mathcal{A}}] \leq E_{\mathbf{x}}[c_1 e^{-c_2 \frac{n R^2 \varepsilon_n^2}{\sigma^2}} I_{\mathcal{A}}] \leq c_1 e^{-c_2 \frac{n R^2 \delta_n^2}{\sigma^2}}.$$

Combining the results yields that the universal positive constants c_5, c_6 exist such that

$$P_{\mathbf{x}, \mathbf{w}}(\mathcal{A} \cap \mathcal{E}) \leq c_5 e^{-c_6 \frac{\min\{1, R^2\} n \delta_n^2}{\sigma^2 + C^2}}.$$

Combining both cases and renaming constants (at the cost of ambiguity) yields the claimed bound. $\square$

6.4 Contraction inequalities and symmetrization

We introduce two auxiliary results and a standard symmetrization argument that will be needed for the proof of Theorem 6.2.1:

We first turn to a more general setting: Define for a given function class $\mathcal{F}$ the random variable

$$Z := \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n f(\mathbf{x}_i). \quad (6.4.1)$$

We are interested in bounds on $P(Z \geq t + E[Z])$ for $t \geq 0$. We first introduce a contraction principle, a useful tool from the theory of empirical processes. Specifically, it will allow us to make use of the localized population Rademacher complexity later on in the proof of Theorem 6.2.1 and also in extending Theorem 6.4.2 yield a weaker bound:

We call a function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ *centered* if $\varphi(0) = 0$.

Theorem 6.4.1 (Rademacher contraction inequality). For any set $\mathbb{T} \subset \mathbb{R}^d$ and any family of centered 1-Lipschitz function $\{\varphi_j, j = 1, \dots, n\}$, we have

$$E \left[\sup_{\theta \in \mathbb{T}} \sum_{i=1}^n \varepsilon_i \varphi_i(\theta_i) \right] \leq 2E \left[\sup_{\theta \in \mathbb{T}} \sum_{i=1}^n \varepsilon_i \theta_i \right],$$

where $\varepsilon_1, \dots, \varepsilon_n$ are i.i.d. Rademacher variables.

Proof. See [4, Th. 11.6]. □

The following theorem provides a bound under the assumption that the function class $\mathcal{F}$ is uniformly bounded by a positive constant $b > 0$.

Theorem 6.4.2 (Talagrand concentration for empirical processes). Consider a countable class of functions $\mathcal{F}$ uniformly bounded by b . Then, for all $\delta > 0$, the random variable Z (6.4.1) satisfies the upper tail bound

$$P(Z \geq E[Z] + \delta) \leq \exp \left(\frac{-n\delta^2}{8eE[\Sigma^2] + 4b\delta} \right), \quad (6.4.2)$$

where $\Sigma^2 := \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n f^2(\mathbf{x}_i)$.

Remarks:

- The condition of $\mathcal{F}$ being countable is constraining here, since we want to apply the concentration for the proof of Theorem 6.2.1 and the underlying function class $\mathcal{F}$ is not necessarily countable. For completeness, we are compelled to discuss details about measurability:

There are conditions on $\mathcal{F}$ that we could impose on $\mathcal{F}$ such as separability wrt. the $\|\cdot\|_\infty$ norm or even weaker conditions [18], so that one

could apply 6.4.2 to an uncountable function class, but this issue can be resolved by going back to (2.4.1) and noting that the underlying supremum of expectations is formed over countable function classes for which Theorem 6.4.2 holds.

- Under the additional condition that $\mathcal{F}$ is zero-mean under P , meaning $E[f(X)] = 0$ for all $f \in \mathcal{F}$, we state a weaker bound by further bounding $E[\Sigma^2]$:

$$E[\Sigma^2] \leq \sigma^2 + 8bE \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{x}_i) \right| \right], \quad (6.4.3)$$

where $\sigma^2 := \sup_{f \in \mathcal{F}} E[f^2(\mathbf{x})]$. We prove this using a standard symmetrization argument involving the contraction inequality 6.4.1: By adding zero we obtain

$$\begin{aligned} E[\Sigma^2] &= E \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n f^2(\mathbf{x}_i) \right] \\ &= E \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (f^2(\mathbf{x}_i) - E[f^2(X)]) + E[f^2(X)] \right] \\ &\leq E \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (f^2(\mathbf{x}_i) - E[f^2(X)]) \right] + \sup_{f \in \mathcal{F}} E[f^2(X)], \end{aligned}$$

where the inequality follows, since a supremum of sums is bounded by the sum of the applicable suprema. Since $\sup_{f \in \mathcal{F}} E[f^2(X)] = \sigma^2$, it remains to show

$$E \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (f^2(\mathbf{x}_i) - E[f^2(X)]) \right] \leq 8bE \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{x}_i) \right| \right].$$

Now let $\{\mathbf{y}_i\}_{i=1}^n$ be a sequence of random variables that is independent of $\{\mathbf{x}_i\}_{i=1}^n$ and follows the same distribution and note that $\|f\|_2^2 = E[f^2(X)]$, then

$$\begin{aligned} E_{\mathbf{x}} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (f^2(\mathbf{x}_i) - \|f\|_2^2) \right] &\stackrel{i)}{=} E_{\mathbf{x}} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n f^2(\mathbf{x}_i) - E_{\mathbf{y}} \left[\sum_{i=1}^n f^2(\mathbf{y}_i) \right] \right] \\ &\stackrel{ii)}{=} E_{\mathbf{x}} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} E_{\mathbf{y}} \left[\sum_{i=1}^n (f^2(\mathbf{x}_i) - f^2(\mathbf{y}_i)) \right] \right] \\ &\stackrel{iii)}{\leq} E_{\mathbf{x}, \mathbf{y}} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (f^2(\mathbf{x}_i) - f^2(\mathbf{y}_i)) \right], \end{aligned}$$

where *i*) follows from (6.1.3) the inequality follows, *ii*) by linearity of expectation and *iii*), since the supremum of expectation is bounded by the expectation of the applicable supremum. Introducing an i.i.d. sequence

of Rademacher variables $\varepsilon_1, \dots, \varepsilon_n$ yields

$$\begin{aligned}
E_{\mathbf{x}, \mathbf{y}} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (f^2(\mathbf{x}_i) - f^2(\mathbf{y}_i)) \right] &\stackrel{i)}{=} E_{\mathbf{x}, \mathbf{y}, \varepsilon} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i (f^2(\mathbf{x}_i) - f^2(\mathbf{y}_i)) \right] \\
&\stackrel{ii)}{\leq} 2E_{\mathbf{x}, \varepsilon} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f^2(\mathbf{x}_i) \right] \\
&\stackrel{iii)}{\leq} 4bE_{\mathbf{x}, \varepsilon} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(\mathbf{x}_i) \right],
\end{aligned}$$

where *i*) follows from A.2.1, *ii*) the fact that suprema of sums are bounded by sums of the applicable suprema and that $-\varepsilon_i$ and ε_i are identically distributed and *iii*) by the Rademacher contraction inequality 6.4.1, since $x \mapsto x^2$ is $2b$ -Lipschitz on $[-b, b]$ by the mean value theorem. We further bound

$$\begin{aligned}
E_{\mathbf{x}, \varepsilon} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(\mathbf{x}_i) \right] &\leq E_{\mathbf{x}, \varepsilon} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(\mathbf{x}_i) \right| \right] \\
&\stackrel{i)}{=} E_{\mathbf{x}, \varepsilon} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i (f(\mathbf{x}_i) - E[f(X)]) \right| \right] \\
&\stackrel{ii)}{=} E_{\mathbf{x}, \varepsilon} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i \left(f(\mathbf{x}_i) - E_{\mathbf{y}} \left[\frac{1}{n} \sum_{i=1}^n f(\mathbf{y}_i) \right] \right) \right| \right] \\
&\stackrel{iii)}{=} E_{\mathbf{x}, \mathbf{y}, \varepsilon} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i (f(\mathbf{x}_i) - f(\mathbf{y}_i)) \right| \right] \\
&\stackrel{iv)}{=} E_{\mathbf{x}, \mathbf{y}} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n (f(\mathbf{x}_i) - f(\mathbf{y}_i)) \right| \right] \\
&\stackrel{v)}{\leq} 2E_{\mathbf{x}} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{x}_i) \right| \right],
\end{aligned}$$

which yields the claim. In *i*) we use the fact that $f(X)$ is zero-mean under P , in *ii*) we use the linearity of expectation, in *iii*) we use Jensen's inequality for the modulus function and the fact that the supremum of expected values is bounded by the expected value of the applicable supremum, in *iv*) we use Lemma A.2.1 and finally in *v*) we use the triangle inequality and that the supremum over a sum is bounded by the sum of the applicable suprema.

- If we want to deduce tail-bounds for suprema of the form

$$Z := \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{x}_i) \right|,$$

we can simply apply Theorem 6.4.2 to the expanded function class $\mathcal{F} \cup -\mathcal{F}$, where $-\mathcal{F} := \{-f \mid f \in \mathcal{F}\}$. Note that for suprema of this form, the bound (6.4.3) turns into

$$E[\Sigma^2] \leq \sigma^2 + 8bE[Z] \quad (6.4.4)$$

We are now ready to state the proof of Theorem 6.2.1:

6.4.1 Proof of Theorem 6.2.1 (one-sided uniform law)

Proof. It suffices to prove the claim for functions on the boundary of the δ -ball: $\partial B_2(\delta) := \{f \in \mathcal{F} \mid \|f\|_2 = \delta\}$. Suppose that inequality (6.2.4) is violated by some $g \in \mathcal{F}$ such that $\|g\|_2 > \delta$. Then by the star-shaped property the function $f := \delta g / \|g\|_2$ is a member of $\mathcal{F}$, since $\delta / \|g\|_2 \in [0, 1]$. By absolute homogeneity of the norm, $\|g\|_n^2 < \|g\|_2^2/2$ is equivalent to $\|f\|_n^2 < \|f\|_2^2/2$.

For a function $f \in \partial B_2(\delta)$, it is equivalent to show that

$$\|f\|_n^2 \geq \frac{3}{4}\|f\|_2^2 - \frac{\delta^2}{4}.$$

In order to prove this bound we want to introduce Talagrand's bound for empirical processes (6.4.2). Since $\mathcal{F}$ is not necessarily uniformly bounded a truncation argument is introduced. For $\tau > 0$ define the function $\varphi_\tau : \mathbb{R} \rightarrow \mathbb{R}$

$$\varphi_\tau(u) := \begin{cases} u^2 & \text{if } |u| \leq \tau, \\ \tau^2 & \text{otherwise,} \end{cases}$$

and define $f_\tau(x) := \text{sign}(f(x))\sqrt{\varphi_\tau(f(x))}$. By definition of f_τ the inequality $\|f_\tau\|_n^2 \leq \|f\|_n^2$ holds and therefore by the triangle inequality

$$\|f\|_n^2 \geq \|f_\tau\|_n^2 \geq \|f_\tau\|_2^2 - \sup_{f \in \mathcal{F}} |\|f_\tau\|_n^2 - \|f_\tau\|_2^2|.$$

We now show that a level $\tau > 0$ exists such that

$$\|f_\tau\|_2^2 \geq \frac{3}{4}\|f\|_2^2 \quad (\text{Claim 1})$$

and simultaneously the following tail-bound holds for some positive universal constants c_1, c_2

$$P(Z_n \geq \delta^2/4) \leq c_1 e^{-c_2 \frac{n\delta^2}{c^2}}, \quad \text{where } Z_n := \sup_{f \in \mathcal{F}} |\|f_\tau\|_n^2 - \|f_\tau\|_2^2|. \quad (\text{Claim 2})$$

For this choice of $\tau > 0$ the event $Z_n < \delta^2/4$ implies the second inequality

$$\|f\|_n^2 \geq \|f_\tau\|_2^2 - \sup_{f \in \mathcal{F}} |\|f_\tau\|_n^2 - \|f_\tau\|_2^2| > \frac{3}{4}\|f\|_2^2 - \frac{\delta^2}{4},$$

which yields the claim, since $Z_n \geq \delta^2/4$ holds with probability at least $1 - c_1 e^{-c_2 \frac{n\delta_n^2}{C^2}}$. We now show that such a choice of τ exists:

We show that the first claim holds: Let $I\{|f(\mathbf{x})| \geq \tau\}$ be the indicator function of the event $|f(\mathbf{x})| \geq \tau$, then

$$\|f\|_2^2 - \|f_\tau\|_2^2 \leq E[f^2(\mathbf{x})I\{|f(\mathbf{x})| \geq \tau\}] \leq \sqrt{E[f^4(\mathbf{x})]}\sqrt{P(|f(\mathbf{x})| \geq \tau)},$$

where the last inequality follows from Cauchy-Schwarz inequality. We further bound

$$\begin{aligned} \sqrt{E[f^4(\mathbf{x})]}\sqrt{P(|f(\mathbf{x})| \geq \tau)} &= \sqrt{E[f^4(\mathbf{x})]}\sqrt{P(f^4(\mathbf{x}) \geq \tau^4)} \\ &\leq C\sqrt{E[f^2(\mathbf{x})]}\sqrt{\frac{E[f^4(\mathbf{x})]}{\tau^4}}, \end{aligned}$$

where the inequality follows from the fourth-moment condition (6.2.2) and the Markov inequality. Applying the fourth-moment condition again yields

$$C\sqrt{E[f^2(\mathbf{x})]}\sqrt{\frac{E[f^4(\mathbf{x})]}{\tau^4}} \leq \frac{C^2}{\tau^2}\|f\|_2^2.$$

Putting the pieces together yields

$$\|f\|_2^2 - \|f_\tau\|_2^2 \leq \frac{C^2}{\tau^2}\|f\|_2^2 \Leftrightarrow \left(1 - \frac{C^2}{\tau^2}\right)\|f\|_2^2 \leq \|f_\tau\|_2^2.$$

Choosing $\tau = 2C$ yields

$$\|f\|_2^2 \geq \frac{3}{4}\|f\|_2^2$$

and thereby the claim.

Proof of claim 2: Note that the function class $\{f_\tau \mid f \in \mathcal{F}\}$ is uniformly bounded by τ , so we can use Talagrand's concentration for empirical processes (6.4.2) on Z_n . We first bound $E_{\mathbf{x}}[Z_n]$ and divide this into two steps:

Step 1: We introduce a symmetrization argument to prove an intermediate bound of $E_{\mathbf{x}}[Z_n]$. Given the i.i.d. sequence of random variables $\{\mathbf{x}_i\}_{i=1}^n$ we introduce another sequence of random variables $\{\mathbf{y}_i\}_{i=1}^n$ that are sampled i.i.d. from the underlying distribution P that are independent of the sequence $\{\mathbf{x}_i\}_{i=1}^n$. We furthermore introduce i.i.d. Rademacher variables $\varepsilon_1, \dots, \varepsilon_n$ that

are also independent of both sequences $\{\mathbf{x}_i\}_{i=1}^n$ and $\{\mathbf{y}_i\}_{i=1}^n$, then

$$\begin{aligned}
E_{\mathbf{x}}[Z_n] &= E_{\mathbf{x}} \left[\sup_{f \in \partial B_2(\delta)} \left| \|f_{\tau}\|_n^2 - \|f_{\tau}\|_2^2 \right| \right] \\
&\stackrel{i)}{=} E_{\mathbf{x}} \left[\sup_{f \in \partial B_2(\delta)} \frac{1}{n} \left| \sum_{i=1}^n f_{\tau}^2(\mathbf{x}_i) - E_{\mathbf{y}} \left[\frac{1}{n} \sum_{i=1}^n f_{\tau}^2(\mathbf{y}_i) \right] \right| \right] \\
&\stackrel{ii)}{=} E_{\mathbf{x}} \left[\sup_{f \in \partial B_2(\delta)} \frac{1}{n} \left| E_{\mathbf{y}} \left[\sum_{i=1}^n (f_{\tau}^2(\mathbf{x}_i) - f_{\tau}^2(\mathbf{y}_i)) \right] \right| \right] \\
&\stackrel{iii)}{\leq} E_{\mathbf{x}} \left[\sup_{f \in \partial B_2(\delta)} \frac{1}{n} E_{\mathbf{y}} \left[\left| \sum_{i=1}^n (f_{\tau}^2(\mathbf{x}_i) - f_{\tau}^2(\mathbf{y}_i)) \right| \right] \right] \\
&\stackrel{iv)}{\leq} E_{\mathbf{x}, \mathbf{y}} \left[\sup_{f \in \partial B_2(\delta)} \frac{1}{n} \left| \sum_{i=1}^n (f_{\tau}^2(\mathbf{x}_i) - f_{\tau}^2(\mathbf{y}_i)) \right| \right] \\
&\stackrel{v)}{=} E_{\mathbf{x}, \mathbf{y}, \varepsilon} \left[\sup_{f \in \partial B_2(\delta)} \frac{1}{n} \left| \sum_{i=1}^n \varepsilon_i (f_{\tau}^2(\mathbf{x}_i) - f_{\tau}^2(\mathbf{y}_i)) \right| \right],
\end{aligned}$$

where $i)$ uses (6.1.3), $ii)$ the linearity of expectation, $iii)$, Jensen's inequality applied to $E_{\mathbf{y}}$ and the modulus function, $iv)$ uses independence of the sequences $\{\mathbf{x}_i\}_{i=1}^n$ and $\{\mathbf{y}_i\}_{i=1}^n$ and the fact that the supremum over the expectation of indexed random variables is bounded by the expectation of the supremum over the indexed random variables and $v)$ using the fact that the random variables $f_{\tau}^2(\mathbf{x}_i) - f_{\tau}^2(\mathbf{y}_i)$ and $\varepsilon_i(f_{\tau}^2(\mathbf{x}_i) - f_{\tau}^2(\mathbf{y}_i))$ follow the same distribution A.2.1, since the sequences $\{\mathbf{x}_i\}_{i=1}^n$ and $\{\mathbf{y}_i\}_{i=1}^n$ are sampled independently and identically.

Step 2: We further bound $E[Z_n]$ by utilizing the Ledoux-Talagrand contraction inequality (6.4.1) and the critical inequality (6.2.3):

$$E_{\mathbf{x}, \mathbf{y}, \varepsilon} \left[\sup_{f \in \partial B_2(\delta)} \frac{1}{n} \left| \sum_{i=1}^n \varepsilon_i (f_{\tau}^2(\mathbf{x}_i) - f_{\tau}^2(\mathbf{y}_i)) \right| \right] \leq 2E_{\mathbf{x}, \varepsilon} \left[\sup_{f \in \partial B_2(\delta)} \frac{1}{n} \left| \sum_{i=1}^n \varepsilon_i f_{\tau}^2(\mathbf{x}_i) \right| \right],$$

which follows from the triangle inequality and from the fact that $\sup(U + V) \leq \sup U + \sup V$, where U and V are real-valued variables indexed over the same set and the supremum is taken over said set. We can now use the Ledoux-Talagrand contraction inequality (6.4.1), since $f_{\tau}^2(\mathbf{x}_i) = \varphi_{\tau}(f(\mathbf{x}_i))$ and $\varphi_{\tau}/2\tau$ is a centered 1-Lipschitz function.

This can be seen as follows: $\varphi_{\tau}(0) = 0$ by definition of φ_{τ} , so φ_{τ} is centered. It remains to prove the Lipschitz-property. Let $x, y \in [-\tau, \tau]$, then by the mean value theorem applied to $z \mapsto z^2$ there exists a ξ between x and y such that

$$|\varphi_{\tau}(x) - \varphi_{\tau}(y)| = |x^2 - y^2| = |2\xi| \cdot |x - y| \leq 2\tau|x - y|.$$

We can extend this argument to $\mathbb{R}$: By noting that for $x \notin [-\tau, \tau]$ and $y \in [-\tau, \tau]$ one has $\varphi_{\tau}(x) = \tau^2$ and then applying the mean value theorem as before to the arguments $\text{sign}(x)\tau$ and y and noting $|\text{sign}(x)\tau - y| \leq |x - y|$.

This allows us to utilize the localized population Rademacher-complexity:

$$\begin{aligned}
2E_{\mathbf{x}, \varepsilon} \left[\sup_{f \in \partial B_2(\delta)} \left| \sum_{i=1}^n \varepsilon_i f_\tau^2(\mathbf{x}_i) \right| \right] &\stackrel{i)}{\leq} 8\tau E_{\mathbf{x}, \varepsilon} \left[\sup_{f \in \partial B_2(\delta)} \left| \sum_{i=1}^n \varepsilon_i f(\mathbf{x}_i) \right| \right] \\
&\stackrel{ii)}{\leq} 8\tau \overline{\mathcal{R}}_n(\delta, \mathcal{F}) \\
&\stackrel{iii)}{\leq} 8\tau \frac{\delta^2}{128C},
\end{aligned}$$

where *i)* uses the the Ledoux-Talagrand contraction inequality (6.4.1) as described above and *ii)* the definition of localized Rademacher complexity and *iii)* uses the fact that δ is a solution to the critical inequality (6.2.3). Inserting $\tau = 2C$ yields

$$E_{\mathbf{x}}[Z_n] \leq \frac{1}{8}\delta^2.$$

Having bounded $E_{\mathbf{x}}[Z_n]$ we can now use Talagrand's bound for empirical processes (6.4.2) on Z_n , more specifically we use the possibly weaker bound (6.4.4), for $Z = Z_n$:

We bound σ^2 (see (6.4.4)): By construction $\|f_\tau\|_\infty^2 \leq \tau^2 = 4C^2$ and therefore

$$E \left[(f_\tau^2(\mathbf{x}) - E[f_\tau^2(\mathbf{x})])^2 \right] = \text{Var}(f_\tau^2(\mathbf{x})) \leq E[f_\tau^4(\mathbf{x})] \leq 4C^2 \|f\|_2^2 = 4C^2 \delta^2.$$

We bound b (see (6.4.4) and Theorem (6.4.2)): By the triangle inequality and $\|f_\tau^2\|_\infty \leq 4C^2$

$$|f_\tau^2(\mathbf{x}) - E[f_\tau^2(\mathbf{x})]| \leq |f_\tau^2(\mathbf{x})| + |E[f_\tau^2(\mathbf{x})]| \leq 8C^2.$$

Inserting the bounds on b and σ^2 and $E[Z_n] \leq \delta^2/8$ into Talagrand's concentration for empirical processes (6.4.2) using (6.4.4) yields: There exist positive universal constants c_1, c_2 such that for all $u \geq 0$

$$P(Z_n \geq \frac{1}{8}\delta^2 + u) \leq P(Z_n \geq E[Z_n] + u) \leq c_1 \exp \left(-c_2 \frac{nu^2}{C^2\delta^2 + C^2u} \right). \quad (6.4.5)$$

Setting $u = \delta^2/8$ yields the claim. $\square$

6.5 A refinement of Theorem 6.2.1

Theorem 6.2.1 bounds a multiple of $\|f\|_2$ with $\|f\|_n$. Naturally, we are also interested in the question of whether we can bound $\|f\|_n$ by a multiple of $\|f\|_2$. In this section, we look at possible approaches to yield such bounds.

Motivated by the proof of Theorem 6.2.1, we require the function class $\mathcal{F}$ to satisfy an additional fourth-moment condition on the empirical measure:

$$\|f^2\|_n^2 \leq C^2 \|f\|_n^2 \quad \forall f \in \mathcal{F} \text{ s.t. } \|f\|_2 \leq 1, \quad (6.5.1)$$

where the constant $C > 0$ is taken from the first fourth-moment condition (6.2.2). We discuss the implications of this condition on the empirical measure after formulating and proving the following theorem:

Theorem 6.5.1. Under the conditions of Theorem 6.2.1 and additionally, the fourth-moment condition on the empirical measure (6.5.1), then with the same positive universal constants from Theorem 6.2.1 the following bound holds for all $\delta \in [\delta_n, 1]$

$$2\|f\|_n^2 \geq \|f\|_2^2 \geq \frac{3}{5}\|f\|_n^2 \quad \forall f \in \mathcal{F} \setminus B_2(\delta), \quad (6.5.2)$$

with probability at least $1 - c_1 e^{-c_2 \frac{n\delta^2}{C^2}}$

Remark: As an extension of Theorem 6.2.1 this Theorem not only upper bounds but also lower bounds $\|f\|_2^2$ by a multiple of $\|f\|_n^2$.

Proof. Note that in the proof of Theorem 6.2.1 it is shown that the bound (6.2.4) is implied by the event $\{Z_n \leq \delta_n^2/4\}$ for $\tau = 2C$ and that this event holds with probability at least $1 - c_1 e^{-c_2 \frac{n\delta^2}{C^2}}$. We now show that the event $\{Z_n \leq \delta_n^2/4\}$ implies (6.5.2).

Again it is enough to show the bound (6.5.2) for elements of $\partial B_2(\delta)$ by the same argumentation as before. Since $Z_n \leq \delta^2/4$ already implies $2\|f\|_n^2 \geq \|f\|_2^2$ for all $f \in \partial B_2(\delta)$ it remains to show that $Z_n \leq \delta^2/4$ implies $\|f\|_2^2 \geq 3\|f\|_n^2/5$ for all $f \in \partial B_2(\delta)$, the claim is equivalent to showing

$$\delta^2 \geq \frac{3}{5}\|f\|_n^2 \quad \forall f \in \partial B_2(\delta),$$

since $\|f\|_2 = \delta$ for $f \in \partial B_2(\delta)$. By definition of f_τ the bound $\|f\|_2^2 \geq \|f_\tau\|_2^2$ holds for any $f \in \mathcal{F}$, therefore for any $f \in \partial B_2(\delta)$

$$\delta^2 \geq \|f_\tau\|_2^2 = \|f_\tau\|_n^2 - (\|f_\tau\|_n^2 - \|f_\tau\|_2^2) \geq \|f_\tau\|_n^2 - Z_n.$$

So $Z_n \leq \delta_n^2/4$ implies

$$\delta^2 \geq \|f_\tau\|_n^2 - \frac{\delta_n^2}{4} \quad \forall f \in \partial B_2(\delta_n).$$

We now show that for $\tau = 2C$ the inequality

$$\|f_\tau\|_n^2 \geq \frac{3}{4}\|f\|_n^2 \quad (6.5.3)$$

holds. This would yield the claim, since this implies

$$\delta^2 \geq \frac{3}{4}\|f\|_n^2 - \frac{1}{4}\delta^2 \Leftrightarrow \delta^2 \geq \frac{3}{5}\|f\|_n^2.$$

We denote the empirical measure² as

$$P_n = \frac{1}{n} \sum_{i=1}^n \delta_{\mathbf{x}_i}$$

²Technically this is a random measure, but since all properties of probability measures hold for realizations of $\{\mathbf{x}_i\}_{i=1}^n$ properties of probability measures can be used accordingly

and expected values with respect to this measure as E_n . This yields (argumentation as in the proof of Theorem 6.2.1)

$$\begin{aligned} \|f\|_n^2 - \|f_\tau\|_n^2 &\leq E_n[f^2(x)I\{|f(x)| \geq \tau\}] \stackrel{i)}{\leq} \sqrt{E_n[f^4(x)]} \sqrt{P_n(|f(x)| \geq \tau)} \\ &\stackrel{ii)}{\leq} C \sqrt{E_n[f^2(x)]} \sqrt{\frac{E_n[f^4(x)]}{\tau^4}} \\ &\leq \frac{C^2}{\tau^2} \|f\|_n^2, \end{aligned}$$

where $i)$ follows from Cauchy-Schwarz inequality, $ii)$ from applying the empirical fourth-moment condition (6.5.1) (note $E_n[f^4(\mathbf{x})] = \|f^2\|_n^2$) and Markov inequality. Inserting $\tau = 2C$ yields the claim. $\square$

We revisit both of the fourth-moment conditions (6.5.1) and (6.2.2). Both conditions are satisfied if a constant $b > 0$ exists such that

$$\sup_{f \in \mathcal{F}} \left(\sup_{\mathbf{x} \in \mathcal{X}} |f(\mathbf{x})| \right) \leq b, \quad (6.5.4)$$

if we choose $C = b$ in both conditions by monotonicity and linearity of the expected value. If (6.5.4) holds we call $\mathcal{F}$ *uniformly bounded* by b or *b -uniformly bounded*. So in the case of a b -uniformly bounded star-shaped function class we can apply Theorem 6.5.1 by choosing $C = b$ in both fourth-moment conditions. In the proof of Theorem 6.5.1, we deal with the truncated function f_τ , where $\tau = 2C$. So if the function class $\mathcal{F}$ is uniformly bounded by b , then this yields $\tau = 2b$. However, this means $f_\tau = f$, which makes the truncation argument in Theorem 6.5.1 redundant. The reason the truncated function was introduced in the proof, was to utilize Talagrand's concentration for empirical processes 6.4.2. In the case of b -uniformly bounded function classes we can use Talagrand's concentration 6.4.2 directly and yield sharper constants:

Theorem 6.5.2. Under the conditions of Theorem 6.2.1 and additionally, $\mathcal{F}$ is uniformly bounded by b , then with the same positive universal constants from Theorem 6.2.1 (choosing $C = b$) the following bound holds for all $\delta \in [\delta_n, 1]$

$$\frac{4}{3} \|f\|_n^2 \geq \|f\|_2^2 \geq \frac{4}{5} \|f\|_n^2 \quad \forall f \in \mathcal{F} \setminus B_2(\delta_n), \quad (6.5.5)$$

with probability at least $1 - c_1 e^{-c_2 \frac{n\delta^2}{b^2}}$

Remark: The Theorem essentially shows that if the function class is b -uniformly bounded we can yield the result as in Theorem 6.5.1 but with sharper constants.

Proof. We note again that it is sufficient to prove the bound uniformly on $\partial B_2(\delta)$. Proving (6.5.5) is equivalent to showing

$$\frac{4}{3} \|f\|_n^2 \geq \delta^2 \geq \frac{4}{5} \|f\|_n^2 \quad \forall f \in \partial B_2(\delta).$$

We define the random variable

$$Z_n := \sup_{f \in \partial B_2(\delta)} |\|f\|_n^2 - \|f\|_2^2|.$$

By definition of Z_n inequalities

$$\begin{aligned} \|f\|_n^2 &\geq \|f\|_2^2 - |\|f\|_2^2 - \|f\|_n^2| \geq \|f\|_2^2 - Z_n \\ \|f\|_2^2 &\geq \|f\|_n^2 - |\|f\|_n^2 - \|f\|_2^2| \geq \|f\|_n^2 - Z_n \end{aligned}$$

hold for all $f \in \partial B_2(\delta)$. Since $\|f\|_2 = \delta$ for all $f \in \partial B_2(\delta)$ this translates to

$$\begin{aligned} \|f\|_n^2 &\geq \delta^2 - Z_n \quad \forall f \in \partial B_2(\delta), \\ \delta^2 &\geq \|f\|_n^2 - Z_n \quad \forall f \in \partial B_2(\delta). \end{aligned}$$

The event $Z_n \leq \delta^2/4$ implies

$$\|f\|_n^2 \geq \delta^2 - \frac{1}{4}\delta^2 \quad \text{and} \quad \delta^2 \geq \|f\|_n^2 - \frac{1}{4}\delta^2,$$

rearranging yields

$$\frac{4}{3}\|f\|_n^2 \geq \delta^2 \quad \text{and} \quad \delta^2 \geq \frac{4}{5}\|f\|_n^2,$$

which would yield the claim. It remains to show $P(Z_n \leq \delta^2/4) \geq 1 - c_1 e^{-c_2 \frac{n\delta^2}{b^2}}$. This bound holds since Z_n is the same random variable as

$$Z'_n := \sup_{f \in \partial B_2(\delta)} |\|f_\tau\|_n^2 - \|f_\tau\|_2^2|$$

for $\tau = 2b$ and we already proved $P(Z'_n \leq \delta^2/4) \geq 1 - c_1 e^{-c_2 \frac{n\delta^2}{b^2}}$ there, note that we choose $C = b$ here. $\square$

6.6 Determining rates

Again the question is posed how one can use the results at hand to determine rates for specific choices of function classes. To apply the one-sided uniform law 6.2.1 the solution δ_n to the critical inequality (6.2.3) is needed. One can estimate a solution δ_n by solving the critical inequality with respect to the localized empirical Rademacher complexity wrt. the function class $\mathcal{H}$

$$\widehat{\mathcal{R}}_n(\delta; \mathcal{H}) := E_\epsilon \left[\sup_{f \in \mathcal{H}} \sum_{i=1}^n \epsilon_i f(\mathbf{x}_i) \right].$$

Here explicit solutions can be calculated using entropic techniques again [20, Cor. 14.3]: Given an arbitrary constant $c > 0$, then any δ that satisfies the inequality

$$\frac{64}{\sqrt{n}} \int_0^\delta \sqrt{\log N(t; B_n(\delta, \mathcal{H}), \|\cdot\|_n)} dt \leq \frac{c\delta^2}{2} \quad (6.6.1)$$

also satisfies the critical inequality

$$\frac{\widehat{\mathcal{R}}_n(\delta; \mathcal{H})}{\delta} \leq c\delta. \quad (6.6.2)$$

If the function class is uniformly bounded one can even make a statement the quality of the estimate [20, Prop. 14.25] and could include this in the one-sided uniform law 6.2.1. If the function class is not uniformly bounded a different approach is needed. Note that the following connection holds between the localized empirical and population Rademacher complexity

$$E_{\mathbf{x}}[\widehat{\mathcal{R}}_n(\delta; \mathcal{H})] = \overline{\mathcal{R}}_n(\delta, \mathcal{H}).$$

Given this connection, if one can find a data-independent solution δ_n for (6.6.2), then subsequently it also solves the associated critical inequality on the population level. We exemplify this by extending Example 5.3.1:

Example 6.6.1. We want to apply Corollary 6.3.1 (again we drop the assumption on $B_{\mathcal{F}^*}(3)$ and assume $R = 1$). The underlying function class is

$$\mathcal{F}_{\text{lin}} = \{f_{\theta}(\cdot) = \langle \cdot, \theta \rangle \mid \theta \in \mathbb{R}^d\}.$$

Since we assume $f^* \in \mathcal{F}_{\text{lin}}$ we obtain $\mathcal{F}_{\text{lin}}^* = \mathcal{F}_{\text{lin}} - f^* = \mathcal{F}_{\text{lin}}$. We already derived data-dependent solutions ε_n to the critical inequality

$$\frac{\mathcal{G}(\varepsilon; \mathcal{F}_{\text{lin}})}{\varepsilon} \leq \frac{\varepsilon}{2\sigma},$$

which take the form $\varepsilon_n^2 = c\sigma^2 r/n$, where c is a positive universal constant and $r = \text{rank}(\mathbf{X})$, where $\mathbf{X} \in \mathbb{R}^{n \times d}$ is the design matrix. Note that ε_n is data-dependent, since r depends on the covariates, however, we can find a data independent solution, namely $\varepsilon_n^2 = c\sigma^2 d/n$ by Lemma 4.1.2 and since $r \leq d$. We now want to find a solution to the critical inequality (6.6.2) for $c = 1/(128C)$ using (6.6.1). Analogous argumentation to Example 5.3.1 yields that there exists a positive universal constant c' such that

$$\delta_n^2 = c'C^2 \frac{r}{n}$$

is a solution to (6.6.2). Note this is still data-dependent via r but we can find a data-independent solution using Lemma 4.1.2 and the fact that $r \leq d$, namely

$$\delta_n^2 = c'C^2 \frac{d}{n}.$$

Inserting this into Corollary 6.3.1 yields: There exist positive universal constants c_0, c_1, c_2 such that

$$\|\widehat{f} - f^*\|_n^2 \leq c_0 \left((C^2 + \sigma^2) \frac{d}{n} + \lambda_n \right)$$

with probability at least $1 - c_1 e^{-c_2 \frac{d^2}{n(\sigma^2 + C^2)}}$.

Chapter 7

Conclusion

Bounding the prediction error and excess risk for a regression problem is essential to assessing its performance. In this thesis, we presented a bound on the prediction error for least-squares estimator with (squared) semi-norm regularization for convex function classes, as well as a one-sided uniform law that allows transitioning from prediction error bounds to excess risk bounds. We have demonstrated that the problem of bounding prediction error and excess risk can be effectively reduced to finding tail bounds on empirical processes for this type of regularization. This methodology offers a framework that can be extended to a variety of other loss functions [20, Th. 13.13 & 14.20]. Furthermore, the methodology also accommodates non-Gaussian noise [20, p.424]. These results underscore the versatility and power of concentration inequalities in the analysis of prediction error and risk in statistical learning. The bound on the prediction error is particularly useful in that it allows the calculation of optimal rates for parametric and non-parametric models [20, ch. 13& 15]. Additionally, by leveraging a one-sided uniform law, we can directly translate prediction error bounds into excess risk bounds.

There are a number of points from which this thesis can be picked up: The prediction error for the regularized least squares estimator essentially shows that for increasing regularizing parameter λ_n the estimation error grows linearly. It remains to discuss procedures by which the regularization parameter λ_n should be chosen to yield favourable performance [6, 12, 14]. Another constraint in both the bounds for prediction error and excess risk is the assumption of star-shapedness or convexity of underlying function classes, which are both crucial since they allow using local complexity measures. Neural networks for popular choices of activation functions do not fulfill the star-shaped or convexity property and also classification problems are not covered. In [13] a learning framework is presented in which prediction error and risk bounds are derived for binary classification which can be extended to more general function classes [13, ch. 3 & 4]. A drawback from our derived excess risk bound is the assumption that the regression function is a member of the underlying function class, which is not always the case in applications, so a different approach which

would include an approximation error is needed. Finally, the refinements of the one-sided uniform law suggest a new application: identifying constraints on function classes such that excess risk and prediction error are of the same order.

Appendix A

Auxiliary Results

A.1 Proof of Corollary 5.2.2

Proof. By definition of $\hat{f}$ and since $f^* \in \mathcal{F}$ the following inequality holds:

$$\frac{1}{2n} \sum_{i=1}^n (y_i - \hat{f}(\mathbf{x}_i))^2 + \lambda_n \|\hat{f}\|_{\mathcal{F}}^2 \leq \frac{1}{2n} \sum_{i=1}^n (y_i - f^*(\mathbf{x}_i))^2 + \lambda_n \|f^*\|_{\mathcal{F}}^2.$$

Inserting $y_i = f^*(\mathbf{x}_i) + \sigma w_i$, applying the binomial formula and cancelling terms of the form $\sum_{i=1}^n (\sigma w_i)^2 / n$ yields

$$\frac{1}{2} \|\hat{f} - f^*\|_n^2 - \frac{\sigma}{n} \sum_{i=1}^n (\hat{f} - f^*)(\mathbf{x}_i) w_i + \lambda_n \|\hat{f}\|_{\mathcal{F}}^2 \leq \lambda_n \|f^*\|_{\mathcal{F}}^2.$$

Rearranging yields

$$\frac{1}{2} \|\hat{f} - f^*\|_n^2 \leq \frac{\sigma}{n} \left| \sum_{i=1}^n (\hat{f} - f^*)(\mathbf{x}_i) w_i \right| + \lambda_n (\|f^*\|_{\mathcal{F}}^2 - \|\hat{f}\|_{\mathcal{F}}^2).$$

Note that all probability bounds (in the different cases) in the proof of Theorem 5.1.1 were derived from tail bounds of suprema over g

$$\frac{\sigma}{n} \left| \sum_{i=1}^n w_i g(\mathbf{x}_i) \right|,$$

where g is in an appropriate subsets of $\partial \mathcal{F}$. Utilizing analogous arguments by controlling tail-bounds of

$$\frac{\sigma}{n} \left| \sum_{i=1}^n (\hat{f} - f^*)(\mathbf{x}_i) w_i \right| = \frac{\sigma}{n} \left| \sum_{i=1}^n g(\mathbf{x}_i) w_i \right| \quad \text{with } g = \hat{f} - f^*,$$

where the suprema are over g in an appropriate subsets of $\mathcal{F}^*$, yields the claim. The bound on the expected prediction error $E[\|\hat{f} - f^*\|_n^2]$ follows by the same argument as in the proof of Corollary 5.2.1. $\square$

A.2 Symmetry argumentent in Chapter 6

We call a real-valued random variable X symmetric if X and $-X$ follow the same distribution:

Lemma A.2.1. Let X be a symmetric random variable and ε be an independent Rademacher variable. Then X and εX follow the same distribution.

Proof. Let A be any (measurable) subset of $\mathbb{R}$, then

$$\begin{aligned} P(\varepsilon X \in A) &= P(\varepsilon X \in A, \varepsilon = 1) + P(\varepsilon X \in A, \varepsilon = -1) \\ &= \frac{1}{2}P(X \in A) + \frac{1}{2}P(-X \in A) \\ &= P(X \in A), \end{aligned}$$

where we use σ -additivity of the probability measure, independence of X and ε and the fact that X and $-X$ follow the same distribution in the consecutive equations. $\square$

Bibliography

- [1] Martin Anthony and Peter L. Bartlett. *Neural network learning: theoretical foundations*. Cambridge University Press, Cambridge, 1999.
- [2] Pierre C. Bellec and Alexandre Tsybakov. Bounds on the prediction error of penalized least squares estimators with convex penalty. *Modern Problems of Stochastic Analysis and Statistics*, pages 315—33, 2017.
- [3] Patrick Billingsley. *Probability and Measure*. John Wiley and Sons, third edition, 1995.
- [4] Stéphane Boucheron, Gábor Lugosi, and Pascal Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, 2013.
- [5] Peter Bühlmann and Sara van de Geer. *Statistics for high-dimensional data*. Springer Series in Statistics. Springer, Heidelberg, 2011.
- [6] Michael Chichignoud, Johannes Lederer, and Martin J. Wainwright. A practical scheme and fast algorithm to tune the lasso with optimality guarantees. *Journal of Machine Learning Research*, 17(229):1–20, 2016.
- [7] Arnak S. Dalalyan, Mohamed Hebiri, and Johannes Lederer. On the prediction performance of the Lasso. *Bernoulli*, 23(1):552–581, 2017.
- [8] Rick Durrett. *Probability: Theory and Examples*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 5 edition, 2019.
- [9] L. S. Nelson F. C. Leone and R. B. Nottingham. The folded normal distribution. *Technometrics*, 3(4):543–550, 1961.
- [10] Trevor Hastie, Robert Tibshirani, and Martin Wainwright. *Statistical learning with sparsity*, volume 143 of *Monographs on Statistics and Applied Probability*. CRC Press, Boca Raton, FL, 2015.
- [11] Johannes Lederer. *Fundamentals of High-Dimensional Statistics: With Exercises and R Labs*. Springer Texts in Statistics, 2022.

- [12] Wei Li and Johannes Lederer. Tuning parameter calibration for ℓ_1 -regularized logistic regression. *Journal of Statistical Planning and Inference*, 202:80–98, 2019.
- [13] Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar. *Foundations of Machine Learning*. MIT Press, second edition, 2018.
- [14] Mahsa Taheri, Néhémy Lim, and Johannes Lederer. Balancing statistical and computational precision: A general theory and applications to sparse regression. *IEEE Transactions on Information Theory*, 69(1):316–333, 2023.
- [15] Mahsa Taheri, Fang Xie, and Johannes Lederer. Statistical guarantees for regularized neural networks. *Neural Network*, 142:148–161, 2020.
- [16] Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B*, 58(1):267–288, 1996.
- [17] Sara A. van de Geer. *Empirical Processes in M-Estimation*. Cambridge University Press, 2000.
- [18] A. W. van der Vaart and Jon A. Wellner. *Weak convergence and empirical processes—with applications to statistics*. Springer Series in Statistics. Springer, Cham, second edition, 2023.
- [19] Roman Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 2018.
- [20] Martin J. Wainwright. *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 2019.
- [21] Larry A. Wasserman. *All of Nonparametric Statistics*. Springer Texts in Statistics, 2006.